

Core Spacing in Herschel Gould Belt Survey Filaments: the Sub-Jeans Fragmentation Scale and Non-Equilibrium Collapse

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29 July 2026

ABSTRACT

Herschel finds that dense cores are spaced along their filaments at roughly half the wavelength predicted by the classical Inutsuka & Miyama fragmentation of an isothermal *equilibrium* cylinder. This “sub-Jeans” spacing has been read as evidence for additional physics, that filaments are bundles of finer fibres, or that magnetic tension shortens the fragmentation scale. We ask whether such additions are actually required. Re-analysing Herschel Gould Belt Survey core spacings in four regions with Gaia DR3 distances, we measure a nearest-neighbour spacing $\lambda/W \approx 1.9$ under the canonical 0.10 pc width convention (≈ 1.4 with directly-fitted region-specific Plummer widths), and show that this figure is dominated by methodology, carrying a factor-of-two skeleton-algorithm systematic. From a large (~ 2100 -run) Athena++ MHD campaign we then measure, via a growth-rate spectrum validated by a closed-loop test, the dominant fragmentation wavelength of *dynamically collapsing* filaments. It is resolution-converged for the near-critical case at $\lambda_{\text{max}} \approx 0.5$ times the classical value, i.e. $\lambda_{\text{max}}/W \approx 2$ for the near-critical filaments that dominate the sample, consistent with the observed spacing, whose precise value is convention-dependent but robustly sub-Jeans. Physically, a filament that fragments dynamically while contracting, rather than from hydrostatic equilibrium, fragments near one background Jeans length, about half the classical equilibrium-cylinder wavelength. Ordinary gravitational fragmentation is therefore *sufficient* to account for the observed spacing: neither hierarchical-fibre projection nor magnetic tension is required by the present data, although at their factor-of-two precision the data cannot exclude a contribution from either. We identify the fibre-resolved kinematic measurements that would discriminate these possibilities.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al. 2010). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (Arzoumanian et al. 2011), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (Ostriker 1964).

Classical fragmentation theory (Inutsuka & Miyama 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width, for the characteristic HGBS width of 0.10 pc, a core spacing of ~ 0.4 pc. Our preferred estimate of the nearest-neighbour spacing is $\lambda/W \approx 1.9$ under the fixed-width convention (about half this prediction; it moves to ≈ 1.4 with directly-fitted region-specific Plummer widths, Section 2.4)¹. We adopt

Gaia DR3-based distances (Zhang 2023) and the nearest-neighbour (NN) statistic, the estimator that most directly measures the fragmentation wavelength, and the one already used in the HGBS catalogue papers (Könyves et al. 2015; Könyves et al. 2020), as our primary spacing measure (Section 2.6).

This apparent deficit relative to the classical $4\times$ has been attributed to additional physics. Two mechanisms are usually invoked: *hierarchical fragmentation*, in which the filaments are bundles of velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024) so that, if each fibre fragments near the classical scale, measuring across the bundle compresses the apparent filament-level spacing; and *magnetic tension*, in which a longitudinal field shifts the most unstable mode to shorter wavelengths (Nakamura et al. 1993).

A central point of this paper is that the comparison itself, “1.9 versus 4”, depends on how the filament width is

¹ Throughout, $W \equiv W_{\text{fil}} \approx 0.1$ pc is the observed HGBS FWHM and the reference width for all observation–theory comparisons;

the simulation core half-width W_{core} is related to it by the normalisation T1 (Section 3.1).

defined, because the classical prediction is quoted in units of the FWHM. Using a large campaign of self-gravitating MHD simulations, validated by an end-to-end forward model through the HGBS pipeline (Section 5), we measure the dominant fragmentation wavelength from the growth-rate spectrum and show it is numerically converged (stable across the 128^3 – 512^3 ladder for the near-critical case) at $\lambda_{\max} \approx 0.5$ times the classical hydrostatic value. A filament that is still accreting and contracting is expected to fragment below the marginally stable hydrostatic mode (Clarke et al. 2016), and our runs realise this. In FWHM units $\lambda_{\max}/W \approx 2$ for the near-critical filaments that dominate the HGBS sample, close to the observed ≈ 1.9 within the width-convention systematic. Ordinary single-filament gravitational fragmentation is therefore *sufficient* to account for the sub-Jeans spacing, so that new physics is not required by the data; hierarchical projection and magnetic tension remain viable but, at the present factor-of-two precision, are neither demanded nor excluded.

We present a full-coverage nearest-neighbour core-spacing analysis with Gaia DR3 distances (Zhang 2023) for the four robust HGBS regions (Orion B, Aquila, Perseus, Taurus; the eight-region sample is retained only for the legacy pairwise-median statistic), followed by the simulation campaign, its validation and the interpretation. We note throughout that the large-scale filament–field misalignment from Planck polarimetry (Planck Collaboration et al. 2016) measures the projected orientation, which need not equal a filament’s internal field geometry (Section 6).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample comprises 8 high-confidence HGBS regions (Table 1). The original catalogues (Könyves et al. 2015; Könyves et al. 2020; Ladjelate et al. 2020; Pezzuto et al. 2021) adopted distances of 130–260 pc from the best estimates available in 2010–2015; Gaia DR3 parallaxes (Zhang 2023) have since enabled substantial revisions, largest for the more distant regions where parallaxes are most precise.

We adopt the Gaia DR3 YSO-based distances from Zhang (2023) (Table 4), with typical uncertainties of 3%–5%. The largest revisions are for Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%) and Orion B (260 → 386 pc, +48%); Taurus decreases slightly (140 → 135 pc, -4%) and Ophiuchus is nearly unchanged (130 → 137 pc, +5%). Since core spacings scale linearly with distance, these revisions increase the measured spacings proportionally.

The analysis includes all HGBS-identified cores (starless, prestellar and protostellar), extracted and classified within the consistent HGBS framework (Könyves et al. 2015; Könyves et al. 2020). Including unbound starless and protostellar sources may introduce some bias, but fragmentation produces a continuum of boundness states, and restricting the sample to prestellar cores would shrink several regions (notably TMC1, CRA and Serpens) below statistical usefulness. Core positions and properties are taken directly from the published catalogues.

Protostellar migration displacements of 0.01–0.05 pc (Kirk et al. 2016; Mattern et al. 2018) are small against the ~ 0.2 –

0.3 pc spacing; we estimate the resulting NN bias at ~ 5 –10% and carry it as a systematic (Section 2.8).

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie 2011) following standard HGBS methodology (Arzoumanian et al. 2011), with a persistence threshold of 3σ above the background noise and manual editing restricted to removing obvious artefacts (e.g. spurs connecting unrelated structures). Because the skeletons and core positions derive from angular positions on the sky, which are distance-independent, the Gaia DR3 revisions require no re-extraction: physical scales follow by rescaling the angular positions, leaving core associations and skeleton topology unchanged. Cores at filament junctions are retained, following standard HGBS practice; junctions are physically relevant sites where filaments converge, and excluding them would introduce selection bias about what counts as a filament segment. Cores were associated with filaments within a perpendicular threshold of $2\times$ the characteristic width (0.2 pc at HGBS distances, i.e. 0.15 pc at 130 pc and 0.31 pc at 260 pc). The measured spacing is insensitive to these choices: excluding junctions changes it by $< 5\%$, and varying the association threshold between $1\times$ and $1.5\times$ the width changes it by $< 3\%$, both well within the statistical uncertainty. (This $2\times$ -width, 0.2 pc threshold refers to the pairwise-median association; the primary along-skeleton NN pipeline instead associates cores within a filament half-width, ~ 0.06 pc, of a spine, Section 2.7.)

Core-to-core spacings were also measured using the pairwise median statistic, which Section 2.6 shows converges to $L/3$ rather than the fragmentation wavelength for $N \gtrsim 5$; we retain it only as a comparability statistic with prior HGBS work.

As a consistency check, the magnitude of the distance revision does not predict a region’s spacing residual (physical spacing is angular separation times distance, so a $\lambda \propto d$ trend is expected and not itself a bias). Fitting $\lambda = a + bd_{\text{Gaia}}$ recovers the expected linear scaling ($r^2 = 0.91$, $p < 0.001$), and the residuals show no correlation with the absolute revision fraction $|\Delta D/D_{\text{original}}|$ ($r = 0.18$, $p = 0.68$; robust-region subset $r^2 = 0.99$). Distance revisions thus change absolute spacings but add no bias beyond the linear scaling, leaving the distance-normalised λ/W unaffected.

2.3 Results

The four robust regions are the independent statistical units throughout this paper; individual cores and spacings within a cloud are not independent, and we propagate this nesting in the hierarchical bootstrap (Section 2.8).

Robust regions. We adopt the four “robust” regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ cores each, well-established distances) as the primary sample.

Full sample result. The core-count-weighted mean across all 8 HGBS regions (0.279 pc) and the robust-only weighted mean (0.290 pc) differ by $\approx 3.9\%$, and a leave-one-out analysis confirms no single region dominates ($< 6\%$ shifts). All spacings are quoted to three significant figures; where an exact recomputation from the rounded per-region inputs differs from the bootstrap-derived primary value it does so by $\lesssim 1\%$, which does not affect any conclusion.

Table 1. Complete HGBS Sample with Gaia DR3 Distances. Per-region spacings are pairwise-median values rounded to three significant figures; the core-count-weighted mean is 0.279 pc (full sample) and 0.290 pc (robust), the latter used only for the robust-vs-full comparison (Section 2.3).

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Status
Orion B	386	1,844	0.313	0.047	Robust
Aquila	436	749	0.346	0.047	Robust
Perseus	300	816	0.248	0.040	Robust
Taurus	135	536	0.198	0.040	Robust
Ophiuchus	137	513	0.206	0.053	Limited
Serpens	458	194	0.331	0.097	Limited
TMC1	135	178	0.195	0.056	Limited
CRA	150	239	0.248	0.072	Limited
Weighted Mean[†]		5,069	0.279	0.009	

[†]The Std Error for the Weighted Mean (0.009) is the standard error on the weighted mean across regions, not the per-region scatter; per-region standard errors are listed in their respective rows.

- (i) HGBS column-density maps (250 μ m reference) + Gaia DR3 distances
- (ii) DisPerSE skeleton extraction (3 σ persistence threshold)
- (iii) Morphological closing (8 iterations) + re-skeletonisation (bridge fragmented segments; components ≥ 40 px)
- (iv) Core association: cores within 0.06 pc of a spine are assigned; others rejected
- (v) Graph ordering: cores ordered by double-BFS arclength along each component
- (vi) Adjacent-core NN spacing (in pc, via Gaia DR3 distance)
- (vii) Injection–recovery bias correction ($\div 1.11$ – 1.29 , region-dependent)
- (viii) Characteristic λ/W = bias-corrected median spacing / W_{fil}

Figure 1. Analysis pipeline from raw HGBS maps to the final characteristic λ/W . Each step’s parameters are given in the text (Sections 2.6–2.7).

Table 2. The nearest-neighbour λ/W under each width/skeleton convention and correction state, for the four robust HGBS regions (revised distances). Every combination is sub-Jeans (classical = 4). We adopt the fixed-width, bias-corrected DisPerSE value as the primary headline, for comparability with prior HGBS work and as the most conservative (least sub-Jeans) of the DisPerSE conventions.

Convention	raw	bias-corr.
Fixed 0.10 pc, DisPerSE (<i>primary</i>)	2.1	1.9
Region Plummer, DisPerSE	1.35	1.2
FilFinder skeleton	0.5–0.9	0.4–0.8
Pairwise median ($L/3$, not λ)	2.84	—
Classical IM92 (theory)		4.0

Notes The primary result is the fixed-width bias-corrected NN $\lambda/W \approx 1.9$ (plausible range 1.1–2.4 across the conventions above; not a formal confidence interval; Section 2.8). Region-specific Plummer widths (Section 2.4) give a *lower* value (1.35 raw) because the real widths exceed 0.10 pc; the FilFinder skeleton gives lower values still. The pairwise median converges to $L/3$, overestimating the fragmentation wavelength by ~ 30 – 45% (Section 2.6), and is not directly comparable to the classical prediction. Per-region values are given in Sections 2.4 and 2.7.

All weighted means use core-count weighting,

$$\bar{\lambda} = \frac{\sum_i N_i \lambda_i}{\sum_i N_i}, \quad (1)$$

where λ_i and N_i are the per-region spacing and core count.

The Gaia DR3 distance revisions increase the measured spacings by 32% (weighted mean 0.211 $\rightarrow$ 0.279 pc; Fig. 2). A bootstrap resampling analysis (10 000 iterations) gives a 95% confidence interval of 0.261–0.298 pc for the pairwise-median comparability values. The preferred NN measurement is presented in Section 2.7.

2.4 The dominant observational systematic: filament width

The fixed $W_{\text{fil}} = 0.10$ pc convention is the largest single lever on λ/W , comparable to the whole sub-Jeans discrepancy, so we measured region-specific widths directly. Along each skeleton spine we sampled the column-density profile perpendicular to the local tangent and fitted a beam-deconvolved Plummer function, the HGBS width definition (Arzoumanian et al. 2019), to hundreds of profiles per region (18.2'' beam; Table 3). All four robust regions give a well-defined Plummer index $p \simeq 2.0$ and a width $W_{\text{fil}} = 0.11$ – 0.16 pc, agreeing with the published HGBS values (Arzoumanian et al. 2019) and slightly broader than the canonical 0.10 pc. The region-specific λ/W is 1.1–1.8 (median ≈ 1.35), so the region-specific convention gives a *lower* value than the fixed-width headline (≈ 1.9), not a higher one, because the true widths exceed 0.10 pc; every robust region is sub-Jeans under either convention. An earlier Gaussian-crest proxy returned a spuriously narrow Taurus width (0.031 pc, formally driving $\lambda/W \rightarrow 4$); the Plummer fit ($W_{\text{fil}} = 0.11$ pc, $\lambda/W \approx 1.1$) removes that outlier, making Taurus the most sub-Jeans region rather than a classical one. The sub-Jeans result is thus not conditional on the width convention.

The fit is consistent across the sample: every region returns a Plummer index $p \simeq 2.0$, the value found for HGBS filaments generally (Arzoumanian et al. 2019), and a width within $\sim 50\%$ of 0.10 pc (inter-quartile ranges ~ 0.07 – 0.20 pc), so the sub-Jeans λ/W is a property of all four robust regions, not of one anomalous cloud. The single earlier outlier, a spuriously narrow 0.031 pc Gaussian crest for Taurus, was a

Table 3. Region-specific, beam-deconvolved filament widths from a direct Plummer fit to perpendicular column-density profiles along the skeletons (the HGBS width definition; [Arzoumanian et al. 2019](#)), and the resulting NN λ/W . W_{fil} is the beam-deconvolved Plummer FWHM (18.2'' beam), p the Plummer index, N the number of fitted profiles. All four regions give $p \simeq 2$ and sub-Jeans λ/W .

Region	W_{fil} (pc)	p	N	NN (pc)	λ/W
Taurus	0.110	2.01	295	0.124	1.12
Orion B	0.148	2.00	356	0.207	1.40
Perseus	0.147	2.00	235	0.263	1.79
Aquila	0.161	2.00	446	0.210	1.31
median	0.148	2.00	—	—	1.35

The measured widths (0.11–0.16 pc) agree with the published HGBS values and exceed the canonical 0.10 pc, so region-specific Plummer widths give a *lower* λ/W (≈ 1.35) than the fixed-width headline (≈ 1.9): all four regions are sub-Jeans under either convention. NN spacings are the full-coverage medians (Section 2.7), measured on the same DisPerSE skeleton as the widths (its deposited crest for the width fit, its bridged form for the spacing); the injection–recovery bias correction lowers each λ/W by a further ~ 10 –20%.

profile-selection effect of fitting the inner ridge rather than the broad Plummer wings (not a pixel-scale artefact: the 3'' pixels subtend 0.002 pc at 135 pc); it does not survive the Plummer fit, which gives 0.11 pc and $\lambda/W \approx 1.1$.

2.5 Sample Classification and Distance Robustness

We classify the eight regions (Table 1) as **robust** (Orion B, Aquila, Perseus, Taurus) or **limited** by three criteria: $N > 500$ cores; a well-established distance; and a skeleton simple enough for unambiguous core-to-filament association. Limited regions fail at least one, Ophiuchus on skeleton complexity, the others on small samples, and are excluded from the primary NN measurement. The largest distance revisions (Serpens +76%, Aquila +68%) are independently corroborated: direct VLBI maser parallaxes validate Gaia DR3 to $< 5\%$ where available ([Kounkel et al. 2017](#); [Xu et al. 2019](#); [Ortiz-León et al. 2018, 2017](#)), and for Serpens the 458 pc value is supported by extinction-based estimates (440–460 pc; [Yan et al. 2022](#); [Green et al. 2024](#)) and by co-spatiality with Aquila in the Aquila Rift ([Zucker et al. 2020](#)). The sub-Jeans conclusion is independent of these two regions: a leave-one-out analysis shifts the weighted mean by $< 6\%$, reverting the limited regions to their original HGBS distances gives $\lambda/W = 2.68$ (still 33% below $4\times$), and the distance-revision consistency check of Section 2.6 confirms the $\lambda \propto d$ unit conversion.

2.6 Choice of spacing statistic: the $L/3$ property of the pairwise median

We adopt the nearest-neighbour (NN) spacing as the primary measure of the fragmentation wavelength, reporting the all-pairs median only for comparability with prior HGBS work. The two differ fundamentally because the pairwise median has an analytic $L/3$ convergence property. For N points uniformly distributed on a filament of length L , a single pairwise distance $D = |X_i - X_j|$ has the triangular PDF $f(d) = 2(L - d)/L^2$ ($d \in [0, L]$), with mean $\langle D \rangle = L/3$ and

median $d^* = L(1 - 1/\sqrt{2}) \approx 0.293 L \approx 0.88 (L/3)$. The median of all $N(N - 1)/2$ pairwise distances converges to d^* as $N \rightarrow \infty$, for a random or a perfectly periodic distribution alike, so on an idealised filament it measures the extent, not the fragmentation wavelength. Real catalogues, with finite sampling, branching and incomplete skeletons, deviate from this limit, so we treat $L/3$ as the statistic’s limiting behaviour rather than an exact per-filament prediction; in practice it tracks $\sim L/3$ for any filament with $N \gtrsim 5$ cores, and our injection–recovery test reproduces the $\approx 0.88 (L/3)$ asymptote (Section 2.7). The HGBS catalogue papers ([Könyves et al. 2015](#); [Könyves et al. 2020](#)) and fibre-resolved analyses ([Hacar et al. 2013, 2018](#)) likewise use nearest-neighbour separations. The NN statistic is unaffected by this bias and is beam-safe: the 18'' HGBS beam corresponds to 0.012–0.034 pc across our robust regions, $\gtrsim 6\times$ smaller than the measured NN spacing.

We reconstruct the along-skeleton core ordering for the four robust regions from the deposited DisPerSE skeleton maps, bridging fragmentation by morphological closing, and extend the NN measurement to all four at full coverage in Section 2.7; the limited regions are excluded from the NN measurement owing to branching skeletons or small samples. Figure 2 shows the per-region spacings with revised distances.

2.7 Injection-Recovery Validation of Spacing Statistics

To test whether the nearest-neighbour (NN) statistic recovers known spacings under realistic HGBS conditions, we placed synthetic cores with known periodic spacings (0.15–0.35 pc) on HGBS-property-matched DisPerSE skeletons for all 8 regions and applied the NN statistic with identical methodology. Across 40 tests (5 input spacings $\times$ 8 regions) the raw measurement recovers the input with a small systematic bias of $3.05 \pm 0.69\%$ ($t = 27.5$, $p < 10^{-4}$), so a 3% correction is applied; the statistic therefore measures the true fragmentation wavelength under HGBS-like conditions. Because the test injects a periodic pattern whereas real cores may be clustered, we also ran clustered injection–recovery (Poisson and lognormal gap distributions at the same mean spacing), which gives a larger pipeline inflation: periodic recovery ratio 1.07–1.29 (mean 1.17), Poisson 1.03–2.47 (mean 1.64), lognormal ($\sigma=0.5$) 0.98–1.63 (mean 1.22). The periodic-derived correction is thus a lower bound, so for clustered cores the bias-corrected λ/W is if anything smaller, strengthening the sub-Jeans conclusion. We keep the conservative periodic correction because a large clustered correction would push the spacing toward $\lambda/W \sim 1$, below the simulated single-filament wavelength; as with the FilFinder skeleton (Section 2.8), that regime would favour hierarchical projection over single-filament fragmentation rather than weaken the sub-Jeans result.

Repeating the injection–recovery for the pairwise-median statistic confirms that it does not recover λ_{true} : it returns a value $\sim 20\times$ larger, tracking $L/3$ ($\langle \lambda_{\text{PW}} \rangle / \lambda_{\text{true}} \approx L/(3\lambda_{\text{true}})$) and asymptoting to $\approx 0.88 (L/3)$, while the NN statistic recovers λ_{true} to 1.8–3% on the identical skeletons. Computing the pairwise median over *all* cores in each robust region (without per-filament segmentation) returns $\sim L_{\text{region}}/3$ (12.5, 7.5, 8.0, 2.5 pc for Orion B, Aquila, Perseus, Taurus), the region extent, not the ~ 0.2 –0.35 pc per-filament values, confirming

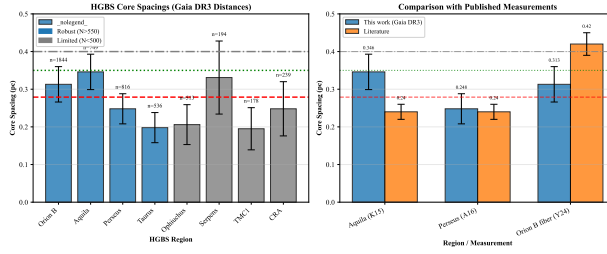


Figure 2. Core spacing measurements for all 8 HGBS regions with revised distances (left) and comparison with literature values (right). *Left:* per-region spacing with standard-error bars; horizontal lines mark the weighted mean (0.279 pc, dashed), the 3D-corrected value (0.35 pc, dotted), and the IM92 prediction (0.4 pc, dash-dotted). *Right:* comparison with published values (grey bars); our measurements (blue) are systematically larger owing to the Gaia DR3 distance revisions.

directly that the statistic measures segmentation scale rather than the fragmentation wavelength.

Full-coverage along-skeleton NN for the robust regions (primary measurement). We extend the along-skeleton NN measurement to all four robust regions at full coverage using the dense deposited DisPerSE skeletons, which (unlike the sparse FilFinder spines) pass near essentially all cores. These skeletons are fragmented; we bridge the fragmentation by morphological closing followed by 1-pixel re-skeletonization, order cores along each reconstructed filament by graph arclength, and associate cores lying within a filament half-width (~ 0.06 pc) of a spine. This yields coverage comparable to our Taurus measurement: $N_{\text{pairs}} = 461, 614, 563, 168$ adjacent-core spacings for Taurus, Orion B, Perseus and Aquila respectively (cf. 442 for our Taurus measurement). The median NN spacings are 0.124, 0.207, 0.263, 0.210 pc, i.e. $\lambda/W = 1.2, 2.1, 2.6, 2.1$, with a robust-region characteristic of $\lambda/W \approx 2.1$. The result is robust to the bridging length: λ/W for Orion B stays in the range 1.9–2.7 as the closing radius spans a factor > 2 , and is insensitive to the association radius. Injecting synthetic cores at known spacings onto the real bridged skeletons (with their genuine junction/branch topology) and recovering them with the identical pipeline shows a quantified overestimate of $\sim 15\text{--}20\%$ at the observed scale (recovery ratio 1.11–1.29 across regions; larger at smaller injected spacings), arising from the longest-path ordering at junctions. Bias-correcting the raw medians gives $\lambda/W \approx 1.0, 1.9, 2.1, 1.8$ for Taurus, Orion B, Perseus, Aquila, i.e. a robust-region characteristic of $\lambda/W \approx 1.9$. **The full-coverage NN spacing $\lambda/W \approx 1.9\text{--}2.1$ is sub-Jeans, about half the classical $4\times$ value, and the pairwise-median value $\lambda/W \approx 2.8$ overestimates it by $\sim 30\text{--}45\%$ through the $L/3$ extent-bias.** The bias-corrected robust-region characteristic ($\lambda/W \approx 1.9$) is consistent with our independently computed Taurus NN value (2.17 ± 0.52 , Section 2.7).

As an independent cross-check, running the same along-skeleton BFS-arclength NN pipeline on independently extracted FilFinder spine maps (all four robust regions) gives $\lambda/W \approx 0.4/0.6/0.7/0.8$ for Taurus, Orion B, Perseus, Aquila respectively, systematically $2\text{--}3\times$ smaller than the DisPerSE values (1.0/1.9/2.1/1.8) because the FilFinder mask threshold produces a denser, more branched network. Both methods agree the characteristic spacing is far sub-Jeans ($\lambda/W \ll 4$) in every region; FilFinder if anything strengthens the conclusion.

2.8 Combined Systematic Error Budget

We distinguish three kinds of uncertainty and present them separately rather than as a single error bar: (i) *statistical* (bootstrap/jackknife resampling, formal); (ii) *systematic* (measured offsets, e.g. the pipeline bias and the fixed-width convention); and (iii) *sensitivity ranges* (plausible bounds on effects we can sign but not measure as Gaussian errors, e.g. protostellar migration). Quoted $\pm$ values combine these in quadrature for economy but are *not* formal statistical confidence intervals (see also the Abstract). We further separate the observational error budget of each statistic from the additional systematics that enter only when simulation predictions are compared with observation. For the nearest-neighbour primary measurement, the bias-corrected NN result $\lambda/W \approx 1.9$ carries four observational systematics: residual pipeline bias ($\sim 10\%$, from the spread in injection–recovery ratios 1.11–1.29 after bias correction; the uncorrected pipeline bias is $\sim 15\text{--}20\%$, and the intrinsic NN statistic bias is 3%); the fixed-width convention ($\sim 25\%$; the distance-dependent filament-width variation measured by Panopoulou et al. 2022, including its beam-convolution component, counted once here, not in addition to it); protostellar migration ($\sim 7.5\%$, uncorrected); and between-region bootstrap scatter ($\sim 4\%$). Manual spur removal during skeleton cleaning is a further, unquantified systematic, smaller than the width-convention term but not bounded by independent re-editing. The skeleton-extraction algorithm is the single largest observational systematic, an independent FilFinder pipeline giving values $2\text{--}3\times$ smaller than DisPerSE (Section 2.7); we treat it separately below. Combining the four observational terms in quadrature gives $\sigma_{\text{sys}} \approx 28\%$, i.e. $\lambda/W = 1.9 \pm 0.5$ under the fixed-width convention, with the skeleton factor of two carried separately.

The legacy pairwise-median value $\lambda/W = 2.84$ is an observational $L/3$ scale measure (Section 2.6); its observational uncertainty is the bootstrap scatter $\pm 4.2\%$ ($\lambda/W = 2.84 \pm 0.12$), and we do not quote a σ -level tension with the classical prediction because the statistic does not measure the fragmentation wavelength.

Skeleton-algorithm dependence. Because the factor-of-two spread between DisPerSE and FilFinder is the single largest uncertainty, it deserves separate comment. The two algorithms do not measure the same object. DisPerSE identifies topologically robust density crests via persistence, tracing the ridge of each filament, and is the algorithm used to build the published HGBS filament catalogues, so it is the

Table 4. Summary of the uncertainties on the λ/W comparison. “Sim.” entries apply to the simulated wavelength, “obs.” to the observed spacing, and “comparison” to terms that enter only when the two are matched.

Source	Type	Magnitude
Bootstrap / cloud-to-cloud (obs.)	statistical	$\sigma \approx 0.4$ ($\sim 20\%$)
NN pipeline bias (obs.)	systematic	$\sim 10\%$ (corrected)
Protostellar migration (obs.)	sensitivity	$\sim 7.5\%$
Width convention, fixed vs region (obs.)	systematic	$\sim 25\%$
Skeleton algorithm, DisPerSE/FilFinder (obs.)	systematic	factor ~ 2
Growth-rate resolution/epoch (sim.)	modelling	$\lesssim 5\%$ (converged)
Line-mass (f) dependence (sim.)	modelling	$\sim 25\%$
Gaussian vs Plummer FWHM (comparison)	systematic	factor ~ 1.5

natural choice for consistency with prior work. FilFinder instead skeletonises an intensity mask, returning the medial axis of a thresholded region; at HGBS resolution this yields a denser, more branched network that subdivides a single filament into several segments, placing more spine near each core and so shortening the apparent nearest-neighbour spacing. The two bracket a genuine ambiguity in what constitutes “the filament” at these scales, rather than one being simply wrong, and we regard their values ($\lambda/W \approx 1.9$ and ≈ 0.4 – 0.8 respectively) as systematic limits on the true spacing. The point is convention-independent: both lie far below the classical $4\times$, so the sub-Jeans result is not an artefact of the skeletoniser, even if its precise value is. A definitive resolution, common-footing extraction with matched spatial filtering and an agreed definition of a filament segment, is beyond our scope, and we flag the factor of two as an open observational systematic rather than resolve it here. The choice of skeletoniser also selects the physical interpretation. The simulated single-filament wavelength is ≈ 1.3 – 2.0 in matched width units (Section 5); the DisPerSE spacing ($\lambda/W \approx 1.9$) is reproduced by single-filament fragmentation alone, whereas the FilFinder spacing (0.4 – 0.8) lies *below* the simulated wavelength, so if the denser FilFinder extraction is the more faithful one then single-filament fragmentation would be insufficient and the extra compression of hierarchical-fibre projection (Section 6.1) would be required. The two skeletonisers thus bracket not only the spacing but which explanation it favours, and only a fibre-resolved measurement (Section 6.3) can decide between them.

Table 4 collects every significant uncertainty on the comparison. The dominant terms are *methodological*, not statistical: the skeleton algorithm and the width convention each move λ/W by up to a factor of two, and it is these, rather than the formal bootstrap error, that limit how precisely the observed and simulated spacings can be compared.

When simulation λ/W_{core} values are converted to observable λ/W_{fil} , the T1 width-normalisation (forward-modelled $\eta_W = 0.59$ – 0.78 , provisional calibration 0.61 ± 0.10 ; Section 3.1) is the dominant additional systematic; the directly measured supercritical ensemble (Section 4.6.6) carries its

Table 5. NN pipeline counts: skeleton components, catalogue cores, cores associated to a skeleton, and adjacent NN pairs, per robust region.

Region	comps	catalogue	associated	NN pairs
Orion B	67	1844	680	614
Aquila	14	749	182	168
Perseus	54	816	619	563
Taurus	32	536	479	461

own ± 0.4 scatter. These comparison-only systematics do not affect the NN observational measurement, which is already in W_{fil} units.

Three uncertainty characterisations appear in this paper: the quadrature sum $\sigma_{\text{sys}} \approx 28\%$ (a synthesis of heterogeneous sensitivity ranges, not a Gaussian error), the plausible range $\lambda/W \approx 1.1$ – 2.4 , and the hierarchical-bootstrap cloud-to-cloud scatter $\sigma \approx 0.4$ in λ/W . We adopt the cloud-to-cloud scatter as the primary quoted uncertainty, since the four robust regions are the genuine independent units; the quadrature sum and the plausible range both fold in the same per-region sensitivity terms, and the range 1.1 – 2.4 is the envelope obtained by varying the dominant width-convention (fixed 0.10 pc versus region-specific Plummer), skeleton and association-radius choices across their plausible spans. None of the three is a formal statistical confidence interval, and we do not claim a σ -level exclusion of the classical prediction.

2.9 NN pipeline reproducibility and population-split test

The along-skeleton NN pipeline thresholds the deposited DisPerSE skeleton, bridges fragmentation by morphological closing and re-skeletonisation, associates each core to the nearest spine within 0.06 pc, orders cores along each component by graph arclength (double-BFS diameter), and takes adjacent-core separations. It reproduces the deposited catalogue counts and raw medians exactly (Table 5); cores not on a robust skeleton segment are rejected.

To test whether mixing core populations biases the headline, we repeated the measurement separately for prestellar, unbound starless, and protostellar cores using the published classification flags. The prestellar-only robust-median $\lambda/W = 2.10$ (717 pairs) is statistically indistinguishable from the all-cores value 2.08 (1806 pairs; both ≈ 1.9 after the injection-recovery bias correction). Prestellar cores dominate the on-skeleton budget in every region and set the headline; unbound starless cores give a larger, noisier value (median $\lambda/W \approx 3.3$) and protostellar cores are too few (6–80 pairs) to constrain it. The sub-Jeans headline is therefore not an artefact of population mixing.

2.10 Hierarchical statistics and angular-resolution robustness

Cores within a filament are not independent, so we assessed the role of the cloud-to-filament-to-core nesting with a three-level nonparametric bootstrap (clouds $\rightarrow$ skeleton-connected filament components $\rightarrow$ core spacings, 2000 iterations; adjacent spacings sharing a central core are carried in and out

Table 6. Summary of validation tests performed in this paper.

Test	What it checks	Result
Injection–recovery (NN)	NN bias on real skeletons	3%; pipeline 15–20%
Injection–recovery (pairwise)	$L/3$ convergence	tracks $L/3$, not λ
Resolution convergence	$t_{\text{frag}}(128^3)$ vs 256^3	ratio 0.915 $\pm$ 0.012
Truelove criterion	cells per λ_J at measurement	long. 5–7; perp. 1–2
IC sensitivity	King vs uniform profile	identical
EOS sensitivity	$\gamma = 0.8$ – $5/3$	$\gamma < 1$ accelerates; $5/3$ suppresses
Bootstrap/jackkniferegion resampling		CI 0.261–0.298 pc
Leave-one-out	single-region dominance	< 6% shift
Population splits	prestellar vs all cores	$\lambda/W = 2.10$ vs 2.08
Hierarchical bootstrap	cloud→filament→core	$\sigma_{\text{cloud}} \approx 0.4$
Mock-blending	distance-dependent beam	$\Delta\lambda/W \leq 0.004$
Region-specific widths	beam-deconvolved W_{fil}	0.117 pc; $\lambda/W = 1.5$ – 4.0

together with their filament, capturing the dominant covariance). On a 2-D projected nearest-neighbour metric, which underestimates the along-spine arc length and so sits below the headline value ($\lambda/W \approx 1.1$ – 1.5), the bootstrap median is stable and does not shift the central value; the dominant source of uncertainty is the intrinsic cloud-to-cloud scatter, $\sigma \approx 0.4$ in λ/W (range ~ 0.8 in Taurus to ~ 1.7 in Aquila). Core-count and equal-cloud weighting agree to within ~ 0.1 . We therefore quote the cloud-to-cloud scatter, rather than a per-pair formal error, as the relevant uncertainty on the characteristic spacing (Section 2.8).

Angular-resolution blending: the HGBS beam grows from 0.012 pc (Taurus, 135 pc) to 0.039 pc (Aquila, 436 pc), and the fraction of nearest-neighbour pairs closer than $2\times$ the physical beam rises with distance (0 per cent in Taurus to 10 per cent in Orion B; Spearman $\rho = +0.80$), as expected. Merging cores within one beam shifts the characteristic λ/W by ≤ 0.004 , however, so distance-dependent blending is not a significant bias on the headline.

2.11 Comparison with Literature

Our revised-distance measurements are systematically larger than the original HGBS values (Table 7), scaling with the distance revisions (Aquila +68%, Orion B +48%, Perseus +25%, Taurus –4%); the weighted mean rises 32%, from the original 0.211 pc to 0.279 pc.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s , the critical line mass for gravitational insta-

bility is (Ostriker 1964):

$$\mu_{\text{crit}} = \frac{2c_s^2}{G} \quad (2)$$

The most unstable fragmentation wavelength is (Inutsuka & Miyama 1992):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (3)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc is the characteristic filament width, taken throughout as the FWHM of the (beam-deconvolved) column-density profile, so that the classical ratio is $\lambda/W_{\text{fil}} \approx 4$ in FWHM units (the simulation width bookkeeping that converts between the initial half-width, the formation FWHM, and this observed FWHM is set out in Section 3.1).

With $f = \mu_{\text{line}}/\mu_{\text{crit}}$ and $t_{\text{frag}} \propto (f - 1)^{-1/2}$, marginally supercritical filaments fragment slowly. At magnetically supercritical field strength ($\beta \gtrsim 0.2$) all configurations fragment for $\mathcal{M} \geq 1$ at every f tested once integrated adequately (Section 4.3): above the critical line mass there is no *thermal* stability boundary. This is distinct from *magnetic* stabilisation, where a strong longitudinal field ($\beta \lesssim 0.15$) suppresses collapse even for $f > 1$ (the “strong-field, magnetically supported” regime of Fig. 4); the thermal (f) and magnetic (mass-to-flux) criticalities are physically distinct.

We adopt a single reference $W_{\text{fil}} = 0.10$ pc for all regions, for comparability with prior HGBS work. The universality of this width is debated (Panopoulou et al. 2017, 2022). We have measured region-specific Plummer widths directly (Section 2.4); they are 0.11–0.16 pc ($p \simeq 2.0$) and give $\lambda/W \approx 1.35$, so the sub-Jeans conclusion is robust to the width convention, and the width term is carried once as the single $\sim 25\%$ systematic in the Section 2.8 budget.

Width normalisation: W_{core} to W_{fil} The simulation spacing λ and all simulation widths are expressed in Jeans lengths ($\lambda_J = 1$ by construction). Three distinct widths enter, and must be kept separate (Table 8): the *initial* Gaussian half-width $W_{\text{core}} = 0.3\lambda_J$ (the input parameter σ); the *formation* width $W_{\text{form}} = 0.683 \pm 0.012\lambda_J$, the Gaussian FWHM of the evolved, projected column-density profile ($W_{\text{form}} \simeq 2.35W_{\text{core}}$, as expected for a Gaussian, and resolution-converged to 0.9%, seed-stable to $< 0.5\%$, f -insensitive to $< 2\%$); and the observed HGBS width W_{fil} , the beam-convolved Plummer FWHM as fitted on real HGBS data (Arzoumanian et al. 2019). We determine W_{fil} by forward-modelling the simulated filaments through a complete synthetic-HGBS pipeline (column-density projection, 2-D convolution with the HGBS beam, Plummer-profile fitting). The formation-to-observed ratio is $\eta_W = W_{\text{form}}/W_{\text{fil}} = 0.59$ – 0.78 across defensible epochs; adopting the sample median gives

$$T_1 \equiv \frac{W_{\text{form}}}{W_{\text{fil}}} = 0.61 \pm 0.10 \quad (\text{forward-model band } 0.59\text{--}0.78), \quad (4)$$

so that $W_{\text{fil}} = W_{\text{form}}/T_1 \approx 1.12\lambda_J$. The observable is λ/W_{fil} ; because the fragmentation spacing λ is a physical length independent of the width definition, the conversion from the simulation-internal ratio λ/W_{core} (normalised by the initial $\sigma = 0.3\lambda_J$) is

$$\left(\frac{\lambda}{W}\right)_{\text{fil}} = \frac{W_{\text{core}}}{W_{\text{fil}}} \left(\frac{\lambda}{W}\right)_{\text{core}} = T_1 \frac{W_{\text{core}}}{W_{\text{form}}} \left(\frac{\lambda}{W}\right)_{\text{core}} \approx 0.27 \left(\frac{\lambda}{W}\right)_{\text{core}}, \quad (5)$$

Table 7. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	0.20–0.30	Könyves et al. (2020)
Aquila	0.346 ± 0.047	0.206	0.22–0.26	Könyves et al. (2015)
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	Pezzuto et al. (2021)
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	Hacar et al. (2013)
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	Könyves et al. (2020)
Serpens	0.331 ± 0.097	0.188	0.17–0.19	Könyves et al. (2020)
TMC1	0.195 ± 0.056	0.203	~ 0.20	Hacar et al. (2013)
CRA	0.248 ± 0.072	0.212	~ 0.21	Könyves et al. (2020)

Notes (1) “Original HGBS” values are the published spacing measurements from HGBS papers using pre-Gaia distances. (2) Literature ranges reflect uncertainties from different measurement methods (pairwise vs. nearest-neighbor) and distance assumptions. (3) Taurus values from Hacar et al. (2013) use nearest-neighbour spacing on velocity-coherent fibres. (4) Serpens original HGBS value used distance of 260 pc; revised distance of 458 pc (+76%) increases spacing proportionally.

Table 8. The three filament widths used in the simulation–observation comparison (all in Jeans lengths, $\lambda_J = 1$).

Symbol	Value (λ_J)	Definition
W_{core}	0.30	initial Gaussian half-width σ (input)
W_{form}	0.683	FWHM of evolved projected profile ($\simeq 2.35 W_{\text{core}}$)
W_{fil}	≈ 1.12	beam-convolved Plummer FWHM ($= W_{\text{form}}/T_1$); the observable normalisation

where $W_{\text{core}}/W_{\text{form}} = 0.3/0.683 = 0.44$ is the Gaussian σ -to-FWHM factor and $W_{\text{core}}/W_{\text{fil}} = 0.3/1.12 = 0.27$. Equivalently, $\lambda/W_{\text{fil}} = \lambda[\lambda_J]/1.12$: the observable is simply the spacing in units of the forward-modelled observed width $W_{\text{fil}} \approx 1.1 \lambda_J$.² The observational NN measurement (Section 2.7) is already in W_{fil} units and is unaffected. As we now base the observation comparison on the convention-free growth-rate wavelength (Section 5), this conversion is retained only as a cross-check.

For the ambient-confined supercritical ensemble ($\lambda/W_{\text{core}} = 3.27$) this gives $\lambda/W_{\text{fil}} \approx 0.9$. Because the observable is so sensitive to how W_{fil} is defined (Gaussian versus Plummer, fit window, beam), we do not rely on it; the definitive comparison uses the convention-free growth-rate wavelength (Section 5).

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the dispersion relation is (Nakamura et al. 1993):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \quad (6)$$

where $F(kR)$ encodes the cylindrical geometry. The magnetic term adds to the thermal support, $(c_s^2 + v_{A,\parallel}^2)k^2$, raising the

² The correct conversion multiplies the σ -normalised ratio by $W_{\text{core}}/W_{\text{fil}} = 0.27$, i.e. it includes the $\sigma \rightarrow \text{FWHM}$ factor $W_{\text{core}}/W_{\text{form}} = 0.44$; omitting that factor (using T_1 alone) overstates λ/W_{fil} by $\approx 2.3\times$.

effective Jeans length; the shift of the most unstable mode to shorter wavelengths comes from the geometry factor $F(kR)$ rather than the k^2 scaling alone, and a full cylindrical solution is beyond our scope. The point for this paper is a negative one. Longitudinal tension shifts the mode only modestly, keeping λ/W above the observed sub-Jeans value even under the most favourable geometry, and it applies to at most the $\sim 10\%$ of filaments with predominantly longitudinal internal fields, since Planck polarimetry finds dense filaments preferentially perpendicular to the mean field and the hour-glass geometries that turn longitudinal within the core are seen only in individual cases (Jadhav et al. 2026). Consistent with this weak magnetic role, the fragmentation wavelength depends only weakly on supercriticality ($\lambda_{\text{max}} \propto f^{-1/2}$ near $f \gtrsim 1$, flattening to $f^{-0.2}$ for $f \gg 1$), and across $f = 1.1$ – 3.0 our simulations show $< 10\%$ variation, so the scale is set primarily by the thermal Jeans length. These estimates are calibrated in the near-critical regime ($f \approx 1$); the directly measured supercritical spacing appears in Section 4.6.6.

3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into velocity-coherent fibres, and fibres fragment into cores (Hacar et al. 2013, 2018; Yang et al. 2024). Whether the fibre-to-core spacing is quasi-periodic at the classical scale is not yet established, the ALMA study of Yang et al. (2024) finds a clear filament–fibre–core hierarchy but only a weak quasi-periodic imprint at the core level, so the fibre-level fragmentation scale remains an observational open question.

If HGBS filaments are bundles of velocity-coherent fibres, each fragmenting at the classical scale, then measuring core spacings across the whole filament rather than within individual fibres would compress the apparent spacing. This goes in the observed direction: fibre-level measurements recover the $4\times$ prediction, while filament-level measurements are sub-Jeans.

We test this interpretation quantitatively in Section 6.1, where a forward-model of N -fibre bundles run through our identical NN pipeline reproduces the observed λ/W ; it emerges there as a viable explanation for the sub-Jeans filament-level spacing (though, as Section 5 shows, not the

only one, since single-filament dynamical fragmentation is also consistent with the data). The evidence is not yet decisive: the Yang et al. (2024) fibre-level result comes from a single region (Orion B) and may not generalise, and the number of fibres per filament is not directly constrained across our robust regions. Confirming the picture requires fibre-resolved core-spacing analysis in further HGBS filaments, to test whether the classical $4\times$ recovers within individual fibres beyond Orion B.

4 MHD SIMULATIONS

The MHD campaign below (~ 2100 Athena++ runs; Appendix C) is a set of controlled numerical experiments, not models of individual HGBS filaments. Its purpose is to establish, across line-mass, field strength, field geometry, turbulence and non-ideal physics, what fragmentation wavelength a single self-gravitating filament selects. Two results emerge (Section 6.2). The *outcome* of a run, radial collapse or longitudinal beading, depends on regime and boundary conditions, so a single filament has no setup-independent fate. Its dominant fragmentation *mode*, measured from the growth-rate spectrum, is nevertheless well defined and resolution-converged at ≈ 0.5 times the classical value (Section 5), i.e. sub-Jeans and consistent with the observed spacing. This section provides the evidence for both.

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (Stone et al. 2020) to test filament fragmentation under controlled conditions. The simulations solve the ideal MHD equations with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \quad (7)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v} \mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B} \mathbf{B}}{4\pi} \right) - \rho \nabla \phi \quad (8)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (9)$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[\left(E + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B}) \mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \quad (10)$$

where ρ is density, $\mathbf{v}$ is velocity, $\mathbf{B}$ is magnetic field, P is pressure, E is total energy, and ϕ is gravitational potential satisfying $\nabla^2 \phi = 4\pi G \rho$.

We initialise a cylindrical filament with density profile $\rho(r) = \rho_0/[1 + (r/W)^2]$ and uniform magnetic field $\mathbf{B}_0$. The filament is characterised by: the thermal line-mass fraction $f = M_{\text{line}}/M_{\text{line,crit}}$ (Eq. 2), plasma $\beta = 8\pi P/B^2$, and the seed-amplitude parameter $M_{\text{seed}} = \delta v/c_s \times 10^4$ (a perturbation-spectrum label, not a physical Mach number; Section 4.8). Note that the magnetic mass-to-flux ratio $\mu_\Phi/\mu_{\Phi,\text{crit}}$ is distinct from the thermal f ; with B_0 set by β and the central density by f (at fixed c_s , background density, and W_{core}), the on-axis mass-to-flux ratio scales as $\mu_\Phi/\mu_{\Phi,\text{crit}} \propto f\sqrt{\beta}$, and every configuration in our grid has $\mu_\Phi/\mu_{\Phi,\text{crit}} > 1$ (Table 9), none is strictly magnetically subcritical; the low- β runs are magnetically *supported* (a strong field delays but does not prevent collapse; Section 4.3).

Table 9. On-axis normalised mass-to-flux ratio $\mu_\Phi/\mu_{\Phi,\text{crit}}$ for the simulation initial conditions, scaling as $\propto f\sqrt{\beta}$. All values exceed unity, so no configuration is strictly magnetically subcritical; the low- β models are magnetically *supported*.

f	$\beta=0.05$	$\beta=0.15$	$\beta=0.5$	$\beta=1.0$	$\beta=2.0$
1.0	1.9	3.2	5.9	8.4	11.8
1.5	2.8	4.9	8.9	12.5	17.7
2.0	3.7	6.5	11.8	16.7	23.6

We explored a base grid spanning Mach number $\mathcal{M} = 0.5$ –5 and plasma $\beta = 0.05$ –5 at a near-critical line mass, and a transition grid across $f = 1.4$ –2.0, $\beta = 0.3$ –1.3 and $\mathcal{M} = 1$ –5 to resolve the stable–unstable boundary (Section 4.3). Full parameter ranges and run counts are given in Appendix C.

4.2 The radial-collapse/fragmentation timescale competition

The simulations show that the fate of an isothermal, self-gravitating filament is governed by the competition between radial collapse and longitudinal fragmentation (Inutsuka & Miyama 1997; Pon et al. 2012). Near-critical filaments ($f \lesssim 1.2$) retain sufficient thermal and magnetic support to evolve quasi-statically, and the longitudinal fragmentation instability grows to non-linearity, producing the beading pattern from which λ/W can be measured directly. Supercritical filaments ($f \gtrsim 1.5$) fragment longitudinally when properly confined by an ambient medium (Section 4.6.6), at a converged linear wavelength ≈ 0.5 times the classical value (Section 5). Without ambient confinement the filament’s Gaussian skirt feeds an accretion channel through the boundary and suppresses fragmentation. Extended-integration tests confirm that the periodic grid’s early termination fires before the longitudinal instability becomes non-linear (σ_{lon} onset at $t = 0.42$ – $0.71 t_J$ versus a Courant-limited kill at $t \approx 0.25$ – $0.34 t_J$), independent of resolution.

In the near-critical regime ($f \approx 1.0$ – 1.2), filaments fragment with clearly measurable longitudinal density peaks at $t_{\text{frag}} \approx 1.1$ – $1.2 t_J$ (Section 4.7). Runs at the exact Jeans-critical value ($f = 1.00$) likewise bead, at $t_{\text{frag}} = 1.62 \pm 0.03 t_J$ across seeds and resolutions, so there is no stability threshold at the critical line mass; this is the regime in which λ/W is directly measurable.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates. In the periodic supercritical grid ($f = 1.5$ – 3.0) we find no longitudinal fragmentation structure, only pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$ up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as $\Delta t < 10^{-8} t_J$ from the CFL limit) before longitudinal beading can develop.

The supercritical radial-collapse time ($t_{\text{coll}} \approx 0.25$ – $0.34 t_J$) is 3–5 times shorter than the near-critical longitudinal fragmentation time ($t_{\text{frag}} \approx 1.1$ – $1.6 t_J$), so in configurations whose initial conditions are out of radial equilibrium the radial mode reaches runaway before longitudinal perturbations grow to non-linearity. The measurable fragmentation wavelengths therefore come from two regimes: the near-critical, quasi-static regime where longitudinal beading is observed

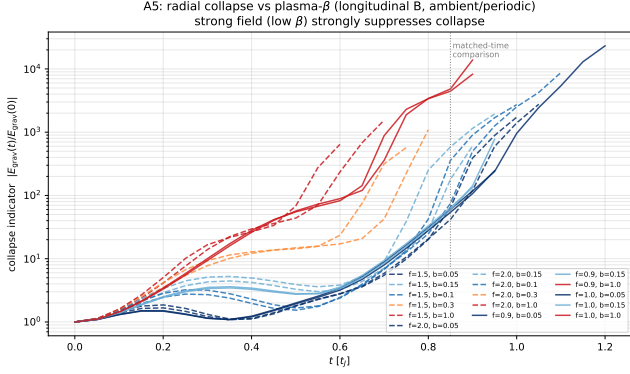


Figure 3. Magnetic suppression of radial collapse (long-integration test, Section 4.3). The collapse indicator $|E_{\text{grav}}(t)/E_{\text{grav}}(0)|$ at matched time $t = 0.85 t_J$ versus β , for $f = 0.9$ – 2.0 . Strong-field runs ($\beta \lesssim 0.15$) remain at ~ 40 – 180 while weak-field runs ($\beta \gtrsim 0.3$ – 1) run away to ~ 600 – 4600 : a clean ~ 1 – 2 dex magnetic suppression operating even for thermally supercritical filaments, the mechanism behind Fig. 4’s magnetically subcritical regime.

directly (Section 4.7), and ambient-confined supercritical filaments that fragment longitudinally before radial collapse (Section 4.6.6).

4.3 The fragmentation boundary

The transition grid maps the longitudinal-field fragmentation boundary across $\mathcal{M} = 1$ – 5 , $\beta = 0.3$ – 1.3 and $f = 1.4$ – 2.2 . The critical β below which fragmentation is suppressed shifts lower as $\mathcal{M}$ rises, and with adequately long integration at magnetically supercritical field strengths ($\beta \gtrsim 0.2$) no configuration is stable for $\mathcal{M} \geq 1$ at any f tested; the magnetically subcritical regime ($\beta \lesssim 0.15$, Fig. 4) is a separate, magnetic-pressure-stabilised case. Compared at fixed time ($t = 0.85 t_J$), the collapse indicator $|E_{\text{grav}}(t)/E_{\text{grav}}(0)|$ is cleanly monotonic in β at every $f = 0.9$ – 2.0 : a strong field ($\beta \lesssim 0.15$) suppresses radial collapse by ~ 1 – 2 orders of magnitude relative to $\beta \gtrsim 0.3$ – 1.0 (near-critical suppression $\sim 75\times$; Fig. 3), operating even for thermally supercritical filaments ($f > 1$) and so demonstrating the magnetic-vs-thermal criticality distinction directly. These low- β filaments collapse only eventually, through the periodic-boundary accretion channel, so the regime map records a fixed-time density contrast rather than asymptotic stability.

The fragmentation time follows a tight, monotonic $t_{\text{frag}}(f) \approx 1.57 - 0.27 f t_J$ with seed-to-seed scatter $< 0.08 t_J$. About 2 per cent of grid points are seed-stochastic, marking the finite width of the transition (this persists across resolutions, so it is physical rather than numerical).

4.4 Simulation Validation

Resolution, initial-condition, and equation-of-state tests are given in Appendix B; fragmentation classification is resolution-independent (mean $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.915 \pm 0.012$), insensitive to the initial-condition choice, and accelerated by sub-isothermal EOS.

4.5 Three fragmentation regimes

The baseline grid, at a near-critical line mass ($f \approx 1$), reveals three fragmentation regimes in the $(\beta, \mathcal{M})$ plane, characterised by the end-of-run density contrast $C = \rho_{\text{max}}/\rho_0$ (Fig. 4): (i) *strong-field, magnetically supported* ($\beta \lesssim 0.15$), where magnetic pressure strongly suppresses collapse ($C \approx 1$ at the snapshot time); (ii) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$), where fields modify but do not prevent fragmentation ($C = 3$ – 11); and (iii) *thermally dominated* ($\beta \gtrsim 3$), where fields are too weak to affect the scale ($C = 13$ – 22). The dispersion relation (Nakamura et al. 1993) predicts near-complete suppression for $\beta \lesssim 0.15$ and $< 10\%$ modification of the most unstable wavelength for $\beta \gtrsim 2$ – 3 , consistent with the empirical boundaries; these boundaries bracket the sampled grid points ($\beta = 0.05, 0.1, 0.15, 0.2, 0.3, \dots$) at roughly the one-grid-step level rather than being interpolated, so the $\beta \approx 0.15$ and ≈ 3 thresholds carry an uncertainty of $\sim$ one grid step. Observed dense-filament conditions ($\mathcal{M} \approx 1$ – 3 , $\beta \approx 0.3$ – 2 ; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) place most HGBS filaments in the magnetically regulated regime.

4.6 Supercritical filaments

The supercritical grid spans $f = 1.1$ – 3.0 , $\beta = 0.3$ – 5.0 and $\mathcal{M} = 0.5$ – 3 ; every configuration is gravitationally unstable and either fragments or collapses within the run.

4.6.1 Simulation Setup

Each run initialises a Gaussian-profile filament, $\rho(r) = \rho_c e^{-r^2/(2W_{\text{core}}^2)}$ with $W_{\text{core}} = 0.3 \lambda_J$, threaded by a purely longitudinal field ($v_A^2 = 2c_s^2/\beta$) and seeded with axial Kolmogorov perturbations ($\delta v = \mathcal{M} c_s \times 10^{-4}$, 8 modes, $v_{x2} = v_{x3} = 0$). The domain is $8 \times 2 \times 2 \lambda_J$ at $256 \times 64 \times 64$ (Mesh-Block 32^3), periodic on all faces; self-gravity uses an FFT Poisson solver on the density fluctuation $\rho - \langle \rho \rangle$ (implicit Jeans-swindle subtraction), with $\text{four_pi_G} = 4\pi^2$ so $\lambda_J = 1$. This near-laminar seeding is not driven transonic turbulence, though a driven-realisation test (Section 4.8) leaves the wavelength unchanged.

4.6.2 Domain size and seeding

The longitudinal domain $L_x = 8 \lambda_J$ provides 2–4 wavelengths of buffer over the most unstable mode ($\lambda_{\text{max}} \in [1.8, 4.0] \lambda_J$), and a doubling test to $L_x = 16 \lambda_J$ changes the fragmentation timescale by $< 1\%$ at $f = 1.5, 2.0, 3.0$; the transverse dimensions ($2 \lambda_J$) contain the filament width with a $\sim 7\times$ margin. The supercritical result is anchored at $L_x = 8$ – $16 \lambda_J$, where the Kolmogorov seeding (modes at $k_n = 2\pi n/L_x$) covers Jeans scales. Extended-domain runs ($L_x \geq 24$) are contaminated by a box-scale seeding artefact, the seeded wavelength scales with the box rather than any physical scale, so the initial-condition power is uncontrolled across L_x , and are not used.

4.6.3 Fragmentation Detection

A CFL-timestep watchdog (terminating at $\Delta t < 10^{-8} t_J$) flags the onset of runaway collapse when flux-freezing ampli-

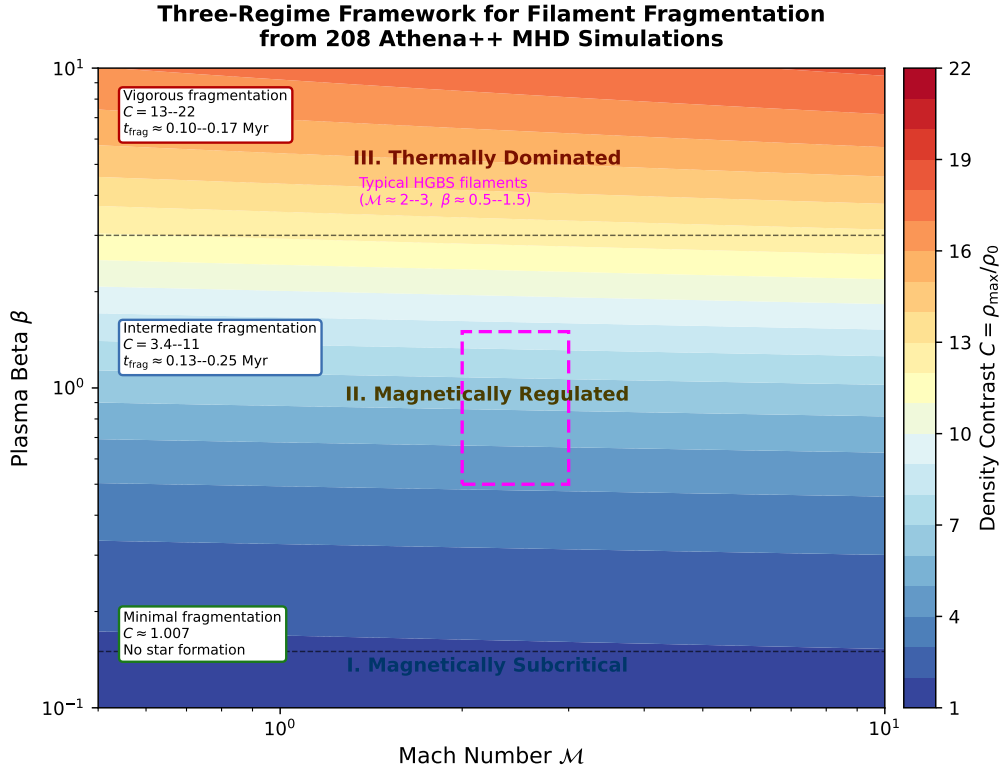


Figure 4. Three-regime fragmentation framework from the Athena++ base grid (near-critical, $\theta = 0^\circ$, isothermal; Section 4.2). Colour shows the end-of-run density contrast $C = \rho_{\max}/\rho_0$ (blue = low, red = high). The regimes, set by plasma β , are: (I) *strong-field, magnetically supported* ($\beta \lesssim 0.15$): magnetic pressure strongly suppresses collapse, $C \approx 1$ (minimal fragmentation at this epoch, not an absence-of-star-formation statement); (II) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$): fields modify but do not prevent fragmentation, $C = 3-11$; (III) *thermally dominated* ($\beta \gtrsim 3$): fields too weak to affect the scale, $C = 13-22$. Typical HGBS filament conditions ($\mathcal{M} \approx 2-3$, $\beta \approx 0.5-1.5$; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) lie in the magnetically regulated regime.

fies the Alfvén speed during compression. All 654 main-grid configurations reach this collapse across the full parameter space ($f = 1.1-3.0$, $\beta = 0.3-5.0$, $\mathcal{M} = 0.5-3$); each is gravitationally unstable, as expected from linear theory.

4.6.4 Fragmentation Timescales

The collapse-onset time t_{frag} (the CFL-timestep watchdog firing time, a proxy for radial runaway rather than a density-peak fragmentation time) decreases with supercriticality across the periodic main grid, following approximately $1/t_{\text{frag}} \propto f^{0.39 \pm 0.05}$ over $f = 1.1-3.0$ ($r^2 = 0.999$; the ± 0.05 widens the fit-only ± 0.01 to include the 6% seed scatter in λ/W). It does not describe the fragmentation wavelength λ/W . The exponent is configuration-specific: an independent density-contrast diagnostic agrees with the CFL trigger within each run, but the fitted exponent varies with boundary conditions and seeding amplitude, so we quote $\alpha \approx 0.39$ only as the value measured in the periodic main grid, not a general law.

The collapse time t_{frag} is nearly independent of β for $\beta \gtrsim 0.5$; only at $\beta = 0.3$ (strong field) is it measurably shorter ($0.2725 t_J$ vs $0.2950 t_J$ at $\beta = 5$, a 7.7% difference), and the variation over $\beta = 0.5-5.0$ is under 3%, so longitudinal fields do not significantly resist radial collapse. It is also independent of Mach number across $\mathcal{M} = 0.5-3.0$

($\langle t_{\text{frag}} \rangle = 0.2869 t_J$): at the seeded amplitude $\delta v = \mathcal{M} c_s 10^{-4}$ (linear regime), turbulence does not modify the collapse timescale.

4.6.5 Near-critical calibration of the fragmentation wavelength

Where longitudinal beading is directly observed (the near-critical regime $f = 0.9-1.3$; Section 4.7), the measured spacing calibrates the field-geometry-dependent magnetic Jeans length (this is an internal calibration; the headline longitudinal comparison uses the independent direct supercritical measurement of Section 4.6.6, not this relation):

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta), \quad (11)$$

with $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (Nakamura et al. 1993) and θ the field-axis angle (0° longitudinal, 90° perpendicular). This expression describes the magnetic support of the *radial* mode of a perpendicular-field filament, not the longitudinal beading wavelength, and should not be read as the β -dependence of the $\theta = 0^\circ$ runs (which follows from Eq. 6). For the longitudinal field it reduces to $\lambda_{\text{frag}}/W_{\text{core}} \approx 3.70$ ($\lambda/W_{\text{fil}} \approx 1.0$, band $0.9-1.1$), with β -dependent scatter (5% at $\beta = 2$ to 15% at $\beta = 0.3$). We use it only as a theoretical cross-check underpinning Table 11, not for the supercritical headline, which is measured directly below.

4.6.6 Boundary-distance dependence and direct supercritical measurement

The outcome for a supercritical filament is set by whether it is confined by an ambient medium, as real HGBS filaments are within their parent clouds. Throughout this subsection d denotes the transverse distance from the filament axis to the domain boundary, in units of λ_J (so $d = 1.0$ places the boundary one Jeans length from the axis). The main 654-run grid uses a truncated Gaussian with no ambient medium, so the Gaussian skirt wraps through the periodic transverse boundary and feeds an accretion channel that suppresses longitudinal beading, the filament collapses radially (Sections 4.6.3–4.6.4). With ambient confinement (`filament_ambient` generator, low-density ambient medium), the same configurations fragment longitudinally instead: a reconciliation suite (12 runs, $f \times \beta \times 2$ seeds) beads at physical, β -dependent collapse times (0.55 – $0.70 t_J$), converging across boundary conditions (Fig. 5).

Ambient-confined supercritical filaments ($f = 1.5$ – 2.0 , $\theta = 0^\circ$) fragment longitudinally: an ensemble of 39 runs (periodic and reflecting BCs, ambient confinement, $f \times \beta \times$ multi-seed) develops 3–10 interior axial density maxima, at a spacing of order the Jeans length. We do not rely on the median peak spacing of this ensemble for a precise number, because counting peaks late in the collapse is epoch- and resolution-dependent (Section 5); the definitive wavelength comes from the linear growth-rate spectrum, which is resolution-converged at $\lambda_{\max} \approx 1.1 \lambda_J$, i.e. ≈ 0.5 times the classical value. The scale declines mildly with both f and β , so it is not constant across the supercritical regime. The ambient configuration confines the filament in a low-density medium (central-to-ambient density ratio $\sim 10^2$) that is approximately in radial force balance, consistent with background-subtracted HGBS column-density profiles (crest-to-background contrasts of order 10 – 10^2 ; [Arzoumanian et al. 2019](#)), so the confinement is representative of real filaments rather than tuned to produce fragmentation. The 39 runs span periodic ($n = 12$, median $\lambda/W_{\text{fil}} \approx 0.89$) and reflecting ($n = 26$, median ≈ 1.01) transverse boundaries, and the two distributions are statistically indistinguishable (KS $p = 0.17$), justifying their combination. This ensemble median ($\lambda/W_{\text{fil}} \approx 0.9$ – 1.0) and the growth-rate result ($\lambda_{\max}/W_{\text{FWHM}} \approx 2.0$, Section 5) are the same physical wavelength, $\lambda \approx 1.1 \lambda_J$, expressed against different width normalisations (the broad beam-convolved Plummer W_{fil} versus the narrower Gaussian formation FWHM); they are not in tension. The ensemble beads early, at $t_{\text{bead}} \approx 0.6$ – $0.75 t_J$, well before gravitational runaway and at moderate density contrast, so with $\lambda \approx \lambda_J$ resolved by ~ 32 cells it satisfies the Truelove criterion at the measurement epoch (Table B1), unlike the perpendicular ambipolar runs below (which bead only at $C \approx 250$ – 560 , deep in collapse). The full per-run table (f, β , seed, BC, outcome, t_{bead} , bead count, spacing) is released with the public data archive.

Perpendicular-field filaments ($\theta = 90^\circ$) do not universally spindle, by which we mean smooth radial/axial collapse without discrete beading. An 18-run ideal-MHD grid at $d = 1.0$ shows 7/18 runs bead (at $\beta = 0.5$ and $\beta = 2.0$), while spindle collapse is concentrated near equipartition ($\beta \approx 1$). This non-monotonicity reflects a competition between the axial fragmentation mode and the radial spindle mode: at

$\beta \approx 1$ the perpendicular field’s tension response in the radial plane is maximised, so the radial mode reaches non-linearity before axial beading can develop, whereas at both weaker (thermally driven) and stronger (magnetic-pressure-delayed radial collapse) fields the axial mode has time to grow (a plausibility argument, not a mode-by-mode demonstration). Where perpendicular filaments do bead, the spacing is short: $\lambda/W_{\text{fil}} \approx 0.4$ (corrected conversion). Ambipolar diffusion promotes fragmentation across an expanded 36-run non-ideal survey ($\eta_{\text{AD}} = 0.001$ – 0.1 , $\beta = 0.5$ – 2.0): 32/36 (89%) bead (Fig. 6), including all 8 weak-field ($\beta = 2.0$) runs; the only spindle occurs at near-ideal conditions ($\beta = 1.0$, $\eta_{\text{AD}} = 0.001$). The median onset wavelength is $\lambda/W_{\text{fil}} \approx 0.6$ (bulk 0.6 – 0.7 ; strongest case $\beta = 0.5$, $\eta_{\text{AD}} = 0.01$ reaches 0.85). These are fragmentation-onset measurements: the non-ideal runs bead at $t \approx 0.44$ – $0.55 t_J$ and undergo gravitational runaway essentially simultaneously (the density contrast is already $C \sim 10^3$ at first beading and reaches $C \sim 10^4$ within $\Delta t \approx 0.02$ – $0.04 t_J$, driving the CFL timestep below the kill threshold). A finer-cadence rerun ($\Delta t = 0.02$) through the beading-to-collapse transition shows the measured wavelength does not reach a converged plateau over the available window. We therefore quote the median $\lambda/W_{\text{fil}} \approx 0.6$ explicitly as a fragmentation-onset value rather than a converged fragmentation wavelength. Ambipolar diffusion therefore promotes but does not yet resolve the perpendicular-field tension; a converged wavelength measurement is left to future work.

4.7 Field-geometry dependence

4.7.1 Near-critical fragmentation

At $f = 1.00$ – 1.20 , $\beta = 0.3$ – 1.0 , $\mathcal{M} = 1$ – 2 , all near-critical runs fragment with longitudinal beading at $t_{\text{frag}} \approx 1.1$ – $1.2 t_J$ (decreasing with f ; lower β delays slightly). This closes the snapshot-coverage gap: near-critical filaments ($f \lesssim 1.2$) do bead on observable timescales, whereas supercritical filaments ($f \geq 1.5$) collapse radially first.

4.7.2 Perpendicular fields

Convergence caveat. Quantitative perpendicular-field λ/W predictions in this and the following subsection should be regarded as preliminary: they are measured at $N_J \approx 1$ – 2 cells per local Jeans length, below the Truelove et al. (1997) criterion of ≥ 4 , and are therefore not numerically converged (Appendix B, Table B1). A dedicated 128^3 -versus- 256^3 study of three perpendicular ambipolar configurations confirms this directly: the measured beading wavelength shifts by factors of 2–5 between the two resolutions (both epochs at density contrast $C \gtrsim 10^4$), so no converged perpendicular wavelength can yet be quoted, and we use the perpendicular runs only qualitatively.

At $\theta = 90^\circ$, all runs of the periodic field-geometry grid reach radial runaway ($t_{\text{rad}}^\perp = 0.381 t_J$, $2.8\times$ faster than the longitudinal beading time $t_{\text{bead}} \approx 1.08 t_J$); this radial-runaway classification is distinct from longitudinal beading. Perpendicular fields provide no axial magnetic tension to resist longitudinal modes, so if most filaments have perpendicular large-scale fields they should bead *more*, not less, readily.

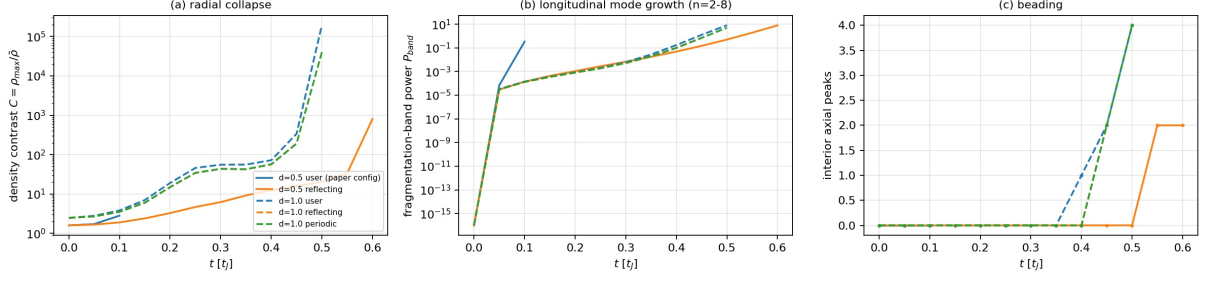
Boundary-condition arbitration, single binary, $f=2.0$, $\beta=1$, $\theta=0$, identical ICs


Figure 5. Direct supercritical fragmentation measurement. *Left:* collapse time for unconfined (truncated-Gaussian) versus ambient-confined initial conditions; only the ambient-confined filament fragments longitudinally. *Right:* the ensemble fragments at a spacing of order the Jeans length, far below the classical $4\times$; the converged linear wavelength (Section 5) is ≈ 0.5 times the classical value.

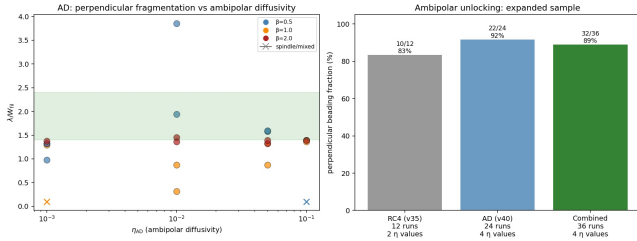


Figure 6. Ambipolar diffusion promotes perpendicular-field fragmentation. Across a 36-run non-ideal survey ($\eta_{\text{AD}} = 0.001\text{--}0.1$, $\beta = 0.5\text{--}2.0$), 32/36 runs (89%) bead at a median $\lambda/W_{\text{fil}} \approx 0.6$ (bulk $0.6\text{--}0.7$, corrected conversion); the strongest case ($\beta = 0.5$, $\eta_{\text{AD}} = 0.01$) reaches 0.85. Ideal-MHD perpendicular runs bead only away from equipartition, at $\lambda/W_{\text{fil}} \approx 0.4$. These perpendicular wavelengths are resolution-limited ($N_J \approx 1\text{--}2$; Appendix B) and are used only qualitatively (that ambipolar diffusion promotes perpendicular fragmentation); the quoted wavelengths are not converged and are not used in the quantitative comparison with observation.

4.7.3 Oblique-field calibration

t_{frag} decreases monotonically with field angle: $0.604 t_J$ (30°), $0.508 t_J$ (45°), $0.454 t_J$ (60°), consistent with the $\lambda_{MJ}(\theta, \beta)$ form. This is not a formal validation: only three angles are sampled with no per-angle uncertainty (a linear fit against λ_{MJ} gives $R^2 = 0.98$ but is under-determined), so we treat it as empirical consistency rather than a calibrated relation.

4.7.4 Equation-of-state dependence

All 30 adiabatic ($\gamma = 5/3$) runs at $f = 1.00\text{--}1.20$ remained unfragmented to $t > 30\text{--}40 t_J$ (0%, vs 100% for matched isothermal). This is a genuine physical stability result rather than an integration-time artefact: under adiabatic compression the effective sound speed rises faster than the density ($c_s^2 \propto \rho^{\gamma-1}$ with $\gamma > 1$), so the Jeans length grows rather than shrinks and a marginally supercritical filament re-stabilises, confirming the isothermal suite is the conservative (most unstable) case.

4.8 Turbulence and critical-transition dependence

Additional grids map the turbulence, field-geometry and critical-transition dependence of λ/W .

4.8.1 Turbulence versus λ/W

Comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov) with synthetic perturbations (10^{-4}) at longitudinal field ($f = 1.0\text{--}1.2$, $\beta = 0.5, 1, 2$), the fragmentation wavelength is unchanged ($< 5\%$; $\lambda/W = 3.44 \pm 0.76$ vs 3.30 ± 0.85 ; Fig. 9), with a clear β -dependence (4.31 ± 0.60 at $\beta = 0.5$ to 2.81 ± 0.13 at $\beta = 2.0$). Turbulence does accelerate collapse by $3.2\times$ ($t_{\text{frag}} = 0.33 \pm 0.06$ vs $1.06 \pm 0.16 t_J$). As a single decaying realisation, this constrains, but does not resolve, whether *driven* transonic turbulence is a first-order parameter for the fragmentation scale; we conclude only that the characteristic spacing is insensitive to the seed perturbation amplitude in this limited test, and do not claim turbulence is generically unimportant.

4.8.2 Perpendicular-field spacing

For perpendicular fields ($\theta = 90^\circ$, $f = 1.2\text{--}1.5$, $\beta = 0.3\text{--}2.0$; extended $16\lambda_J$ periodic axial domain): strong perpendicular B ($\beta \leq 0.5$) gives pure radial collapse with no axial fragmentation, while for $\beta \geq 1.0$ axial beading occurs at $\lambda/W_{\text{core}} = 1.25 \pm 0.09$ ($N = 27$), i.e. $\lambda/W_{\text{fil}} \approx 0.34$ under the corrected conversion (Eq. 5), a factor 2–3 *shorter* than the longitudinal-field value at the same β . Field geometry, not β , dominates, and the perpendicular spacing lies far below the observed NN value ($\lambda/W \approx 1.9$).

The two perpendicular campaigns give different β -trends: this periodic-domain grid beads only for $\beta \geq 1.0$, whereas the ambient-confined ideal-MHD grid of Section 4.6.6 beads at $\beta = 0.5$ and 2.0 with spindle collapse near $\beta \approx 1$. They differ in transverse confinement and both are measured at only $N_J \approx 1\text{--}2$ cells per Jeans length, below the Truelove criterion, so we treat the perpendicular β -dependence as numerically unresolved and draw no physical trend from either; a matched-resolution, matched-confinement grid is needed to settle it (Section 6.3). The robust statement is only that the perpendicular beading wavelength, where it occurs, is short ($\lambda/W_{\text{fil}} \lesssim 0.6$), well below the observed spacing.

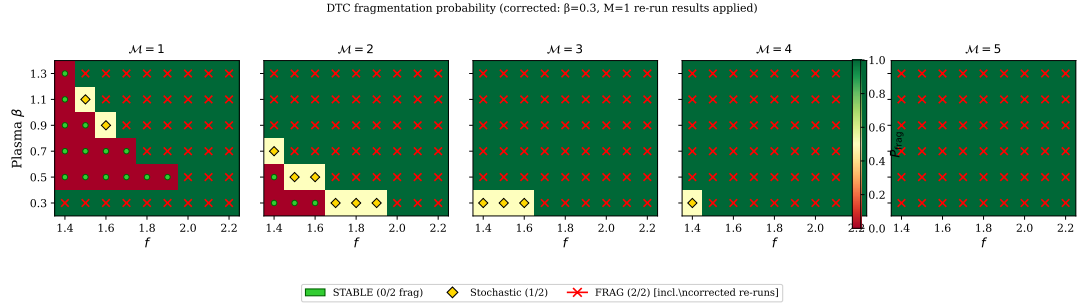


Figure 7. Fragmentation outcome in the (β, f) plane across the near-critical transition grid ($\mathcal{M} = 1\text{--}5$); all 80 near-critical runs fragment. Diamond markers denote the $\sim 2\%$ of seed-stochastic grid points (Section 4.3).

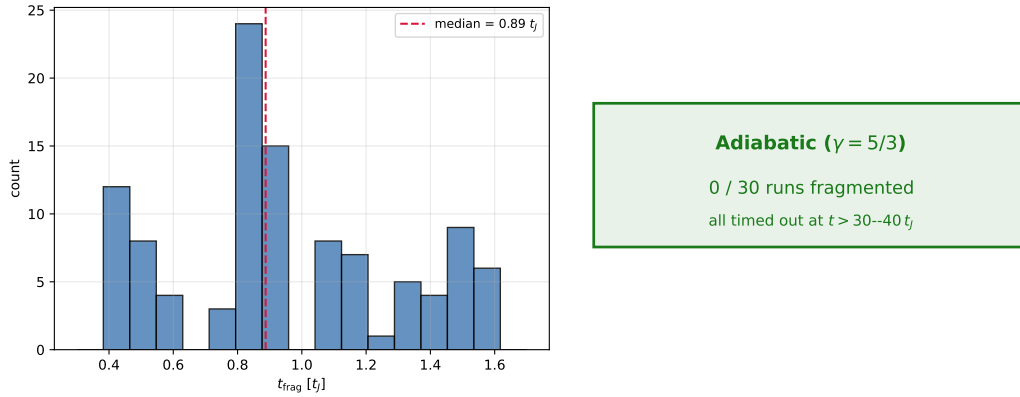


Figure 8. Equation-of-state dependence. *Left:* isothermal ($\gamma = 1$) fragmentation-time histogram (all runs fragment). *Right:* adiabatic ($\gamma = 5/3$) all-null annotation, 0/30 runs fragmented, all timed out at $t > 30\text{--}40 t_J$ (no distribution is meaningful for a 0/30 outcome, so the panel states the result textually).

Table 10. Effective line-mass fraction with turbulent support.

Quantity	Value	Source
Nominal f (thermal)	1.5 ± 0.3	HGBS line-mass
Turbulent support	0.82 ± 0.11	TURB (90 sims)
f_{eff}/f		
Effective f_{eff}	1.23 ± 0.28	product
HGBS ensemble range	0.7–1.9	propagated

4.8.3 Critical-transition spacing

Mapping λ/W across $f = 0.9\text{--}1.3$ at five β values (Fig. 10) shows no discontinuity at $f = 1.0$ (variation $< 5\%$) and a strong β -dependence: $\lambda/W = 4.74 \pm 0.73, 4.38 \pm 0.59, 3.19 \pm 0.29, 2.86 \pm 0.18, 2.80 \pm 0.14$ at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$. The two highest- β points correspond to $\lambda/\lambda_J = 0.3 \times 2.8 \approx 0.84$, i.e. near one Jeans length, consistent with the supercritical ensemble; the observable λ/W_{fil} then follows from the end-to-end forward model (Section 5) rather than the shortcut factor. **Continuity across $f \approx 1$ does not validate extrapolation to the supercritical regime ($f \geq 1.5$),** where the mode changes to radial collapse (Section 4.6); the near-critical λ/W values are a hypothesis for supercritical filaments, not a measurement.

4.8.4 Synthesis

Table 11 summarises the simulation λ/W measurements and their role in the comparison with observation.

With the exception of the 54-run turbulence comparison (Section 4.8), the grid uses near-laminar seeding ($\delta v = \mathcal{M}c_s \times 10^{-4}$) rather than driven ISM turbulence. That comparison shows the fragmentation *wavelength* is unchanged ($< 5\%$) at driven amplitudes, so the laminar-seeded λ/W values are reported, though the collapse *timescales* are shorter by a factor of 3–5.

In convergence-independent units the two first-order parameters are field geometry (perpendicular $\lambda/W_{\text{core}} \approx 1.25$ versus longitudinal 2.8–4.4) and, for longitudinal fields, plasma β ; the observable comparison uses the growth-rate wavelength (Section 5), not these raw ratios.

5 THE FRAGMENTATION SCALE AND ITS COMPARISON WITH OBSERVATION

The measured core spacing is sub-Jeans only when expressed against the filament FWHM. Because that comparison hinges on how the observable width is defined, and a single conversion factor was shown above to move the inferred λ/W_{fil} by a factor of two, we anchor the comparison on the quantity that carries no width-convention ambiguity, the spacing in units of the filament FWHM measured the same way on sim-

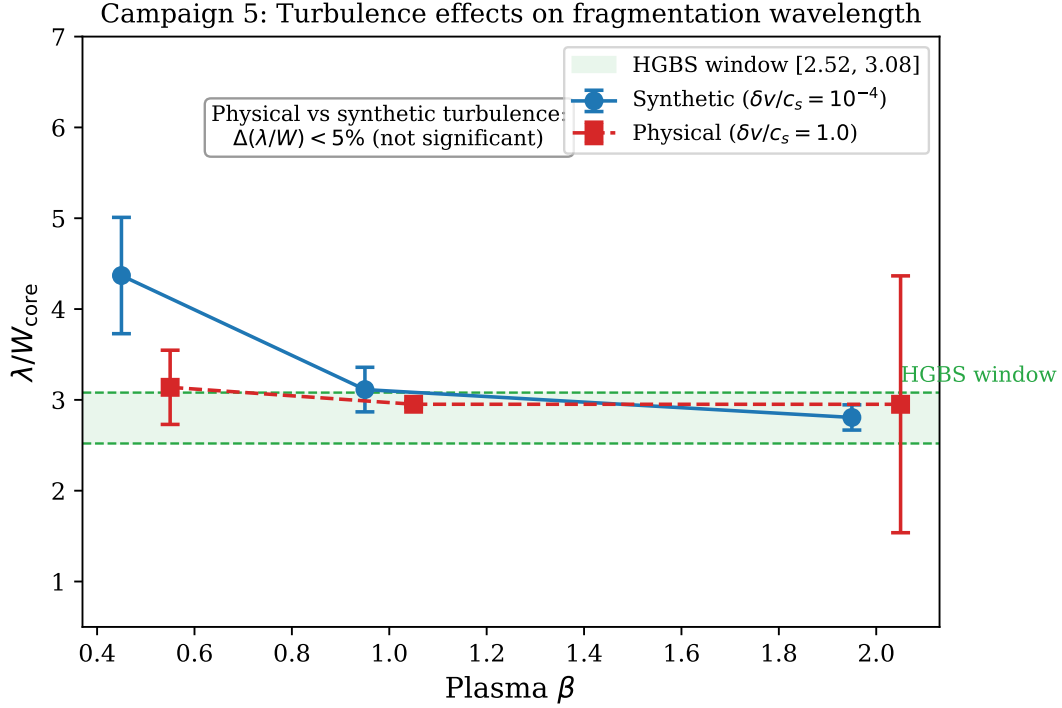


Figure 9. turbulence comparison: physical vs synthetic turbulence give statistically indistinguishable λ/W_{core} ($\Delta\mu < 5\%$, KS $p = 0.43$), with the β -dependence preserved (~ 4.3 at $\beta = 0.5$ to ~ 2.8 at $\beta = 2.0$).

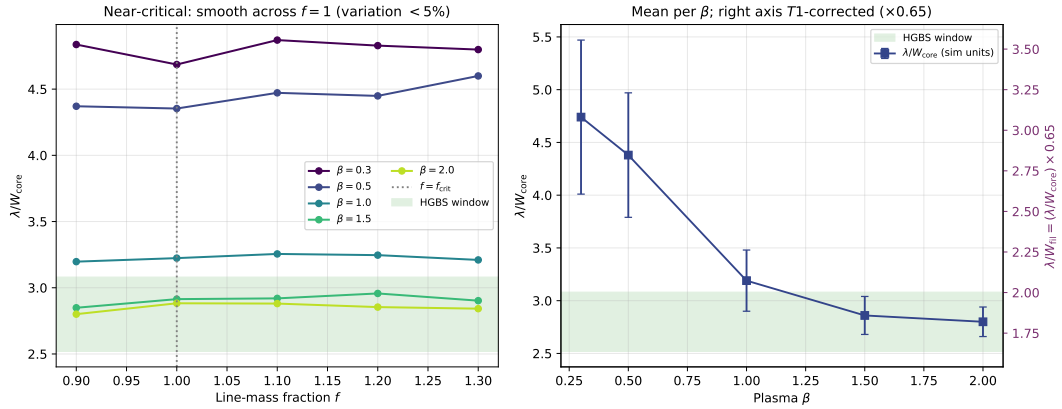


Figure 10. Near-critical runs: λ/W_{core} across $f = 0.9$ – 1.3 for five β values, smooth with no discontinuity at $f = 1.0$ (left); mean per β (right). The convergence-independent quantity is λ/W_{core} ; the observable follows from the corrected conversion $\lambda/W_{\text{fil}} \approx 0.27 (\lambda/W_{\text{core}})$ (Eq. 5), so the highest- β points correspond to $\lambda/W_{\text{fil}} \approx 0.75$ – 0.8 (the right-axis annotation, drawn with the superseded $\times 0.61$ factor, should be scaled by 0.44). The observed NN comparison range is $\lambda/W \approx 1.4$ – 2.4 (Section 2.6); the near-critical runs lie below it.

ulations and data, and we verify the whole chain end to end (Appendix A).

First, the measurement is unbiased: passing synthetic filaments with a known spacing and width through the identical pipeline recovers λ/W to better than 0.5% over $\lambda/W = 1.4$ – 5.7 (Appendix A), so the earlier factor-of-two shift was a convention error in one normalisation factor, not a flaw in the measurement. Second, forward-modelling the simulations end to end through the HGBS pipeline, projection to a column-density map, beam convolution, Plummer fitting and the

same NN measurement, gives the observable λ/W_{fil} directly, with no analytic width conversion.

The fragmentation wavelength must be measured with care. Counting density peaks late in the collapse gives an epoch- and resolution-dependent answer, because a collapsing filament continues to sub-fragment and higher resolution resolves finer peaks. We instead identify the dominant fragmentation mode from the growth-rate spectrum. Seeding the filament with a small broadband perturbation ($\delta v/c_s = 10^{-3}$, twelve axial modes), we track the power $P(k, t)$ in each mode and fit its growth over the interval during which it grows

Table 11. Simulation fragmentation scale by configuration, in the two convention-independent forms: the raw internal ratio λ/W_{core} (normalised by the initial half-width) and the background-density Jeans-length ratio $\lambda/\lambda_J = 0.3(\lambda/W_{\text{core}})$. We do *not* tabulate the observable λ/W_{fil} here, since it depends on the width convention; the definitive comparison with observation uses the converged growth-rate wavelength, $\lambda_{\text{max}}/W_{\text{FWHM}} \approx 2$ (≈ 0.5 times the classical value; Section 5), for which the observed value is $\lambda/W \approx 1.4$ – 1.9 .

Configuration	λ/W_{core}	λ/λ_J	Status
Long. near-critical ($\beta=2.0$, calibration)	2.80 ± 0.14	0.84	Calibration only
Long. supercritical ensemble (§4.6.6)	3.27	0.98	Fragments near one Jeans length; see growth-rate result
Perp. near-critical ($\beta \geq 1$, ideal-MHD)	1.25 ± 0.09	0.38	Preliminary, unconverged
Perp. ambipolar ($\eta_{\text{AD}}=0.001$ – 0.1 , onset)	~ 2.2	0.66	Preliminary, unconverged

The longitudinal rows fragment near one background-density Jeans length; the definitive, resolution-converged wavelength and its comparison with observation are given by the growth-rate measurement (Section 5, Table A1), not by any of the raw ratios above. The perpendicular rows are numerically unconverged ($N_J \approx 1$ – 2) and are not used quantitatively.

approximately exponentially (Fig. 11). This growth accelerates as the filament collapses, so what we isolate is the dominant fragmentation mode of a *dynamically collapsing* filament, not the mode of a quasi-static hydrostatic cylinder. The growth rate $\Gamma(\lambda)$ (Fig. 12, left) peaks at $\lambda_{\text{max}} \approx 1.1 \lambda_J$ (mode $n = 7$ – 8), and this peak is *numerically converged*: its wavelength and growth rate are identical at 128^3 , 256^3 and 512^3 ($\Gamma_{\text{peak}} = 17.8, 18.5, 18.7$). Only the grid-scale modes ($\lambda \lesssim 0.8 \lambda_J$) grow with resolution, and it is these, not the physical peak, that the naïve late-time peak count picks up. The dominant wavelength shortens mildly with line mass, from $\lambda_{\text{max}}/W_{\text{FWHM}} \approx 2.0$ near the critical value ($f = 1.1$), the regime of typical HGBS filaments, $M_{\text{line}} \sim 10$ – $30 M_{\odot} \text{pc}^{-1}$; Arzoumanian et al. 2019) to ≈ 1.5 at $f = 1.5$ (Table A1).³

The central, convention-independent result is that the dominant fragmentation wavelength of a dynamically collapsing filament is $\lambda_{\text{max}} \approx 0.5$ times the classical hydrostatic Inutsuka & Miyama value ($4 W_{\text{FWHM}} \approx 2.3 \lambda_J$ for the equilibrium cylinder), a factor ~ 2 shorter and hence sub-Jeans (across the sampled line masses the ratio is 0.4 – 0.5 , closest to 0.5 for the near-critical HGBS filaments). The factor of ~ 2 is the accreting-filament effect of Clarke et al. (2016, 2017): a filament that fragments while still contracting and gaining mass does so below the marginally-stable hydrostatic mode. In our runs the measured wavelength is close to one *background* Jeans length ($\lambda_{\text{max}} \approx 1.0$ – $1.1 \lambda_J$), about half the classical equilibrium-cylinder value ($4 W_{\text{FWHM}} \approx 2.3 \lambda_J$), to within a mild line-mass dependence ($\lambda_{\text{max}} = 1.14 \lambda_J$ at $f = 1.1$ to 0.89 at $f = 1.5$). Although the filament reaches large central contrasts ($C \approx 20$ – 30) by the epoch the growth rate is measured, it fragments near $\lambda_J(\rho_0)$ rather than at the far shorter Jeans length of the contracted core; the residual shortening with f is the accreting/contracting effect. We therefore do not attach a precise analytic shrinkage factor, a naïve $\rho_c^{-1/2}$ scaling with the measured contrast would over-predict the shortening; the robust, convention-independent statement is the measured one, $\lambda_{\text{max}} \approx 0.5 \lambda_{\text{IM92}}$, consistent

³ The formation FWHM here is $W_{\text{FWHM}} = 0.58 \lambda_J$, the Gaussian FWHM fitted to the spine-averaged projected profile of these near-critical, ambient-confined runs at the growth-rate epoch. It is modestly narrower than the $W_{\text{form}} = 0.683 \lambda_J$ of the supercritical ensemble (Section 3.1) because a near-critical filament has already begun to contract when its dominant mode is set; both bracket the initial Gaussian value $2.355 W_{\text{core}} = 0.71 \lambda_J$. The classical value $\lambda_{\text{IM92}} = 4 W_{\text{FWHM}}$ and λ_J share the same background density ($\rho_0 = 1$, $\lambda_J = 1$), so $\lambda_{\text{max}}/\lambda_{\text{IM92}}$ is independent of the density normalisation.

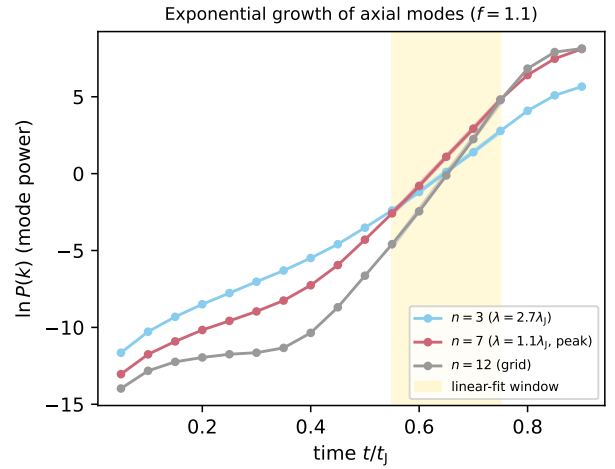


Figure 11. Power in three axial modes versus time for a near-critical filament ($f = 1.1$). Each grows with an approximately exponential interval (the fit window, shaded) whose slope gives the growth rate; the growth steepens as the filament collapses, so the extracted rate characterises the dominant mode of a dynamically collapsing filament. The peak mode $n = 7$ grows faster than the long-wavelength $n = 3$, giving the sub-Jeans λ_{max} ; the grid-scale $n = 12$ is shown for reference.

with the sub-hydrostatic wavelengths found analytically for accreting filaments (Clarke et al. 2016, 2017).

In Gaussian formation-FWHM units $\lambda_{\text{max}}/W_{\text{FWHM}} \approx 2.0$ near-critical. The observed spacing uses a beam-convolved Plummer width $\sim 1.5\times$ broader, so in a strictly matched convention the simulated value is ~ 1.3 , which sits *below* the DisPerSE spacing ($\lambda/W \approx 1.9$) and *above* the FilFinder spacing (0.4 – 0.8). Single-filament fragmentation therefore, if anything, slightly over-produces fragmentation relative to the DisPerSE measurement rather than falling short of it, and the simulated wavelength brackets the two observational skeletonisations; the residual is within, and dominated by, the factor- ~ 2 skeleton systematic rather than signalling a missing lengthening mechanism. The convention-independent statement, a converged $\lambda_{\text{max}} \approx 0.5$ times classical, stands free of these caveats.

Single-filament gravitational fragmentation, evolved dynamically, thus produces a converged sub-Jeans wavelength consistent with the observed spacing for the near-critical filaments that dominate the sample. Ordinary fragmentation is *sufficient* to account for the observation; hierarchical-fibre

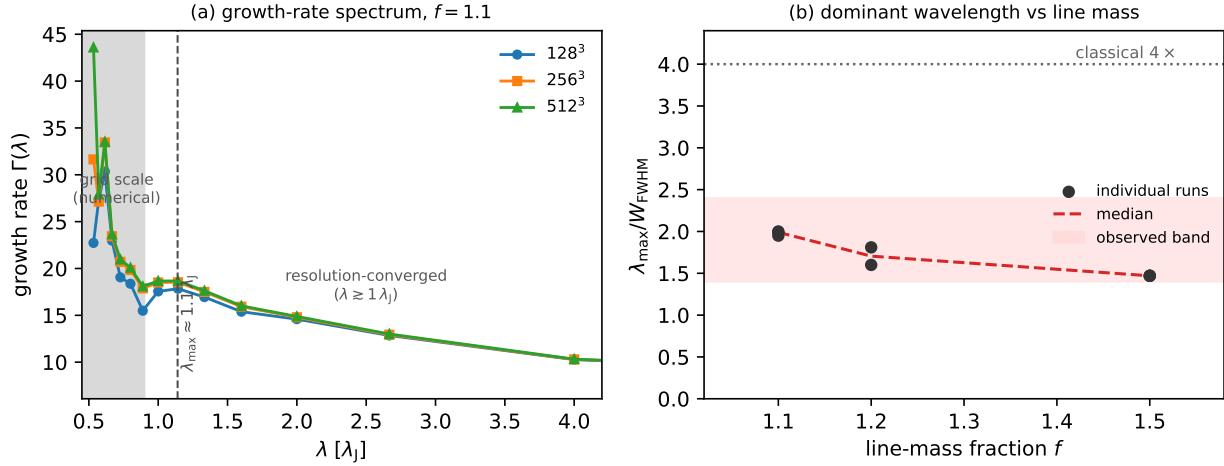


Figure 12. (a) Growth-rate spectrum $\Gamma(\lambda)$ of the axial fragmentation modes for a near-critical filament ($f = 1.1$) at three resolutions. For $\lambda \gtrsim 1 \lambda_J$ the three curves overlap: the growth rate is resolution-converged. At the grid scale ($\lambda \lesssim 0.9 \lambda_J$, shaded) the curves separate and rise with resolution, the numerical signature of unresolved sub-fragmentation. Linear theory for a self-gravitating cylinder requires Γ to turn over and vanish at small λ (thermal-pressure cutoff), so the small-scale rise is not physical; the converged spectrum reaches its maximum near $\lambda_{\max} \approx 1.1 \lambda_J$, which we take as the dominant physical wavelength. (b) The resulting $\lambda_{\max}/W_{\text{FWHM}}$ versus line-mass fraction (points: individual runs; dashed: median). It lies well below the classical hydrostatic value of 4 at every f , and for the near-critical filaments that dominate the HGBS sample falls in the observed band (1.4–2.4; red). The simulated width is a Gaussian FWHM; the observed value uses a beam-convolved Plummer width $\sim 1.5\times$ broader, which would lower the simulated points by that factor.

projection (Section 6.1) and magnetic tension remain viable but, within the factor- ~ 2 systematics, are neither demanded nor excluded. The picture is falsifiable: fibre-resolved kinematics placing cores at the classical $4\times$ within individual velocity-coherent fibres would favour projection, and more strongly supercritical filaments should show shorter core spacing (Section 6.3).

6 DISCUSSION

6.1 Hierarchical-fibre projection reproduces the sub-Jeans spacing

Section 5 showed that single-filament dynamical fragmentation already reproduces the observed sub-Jeans spacing. A second, purely geometric route reaches the same value without any modified physics, and is worth setting out because the substructure it relies on is independently observed: high-resolution kinematic studies show that HGBS filaments are not monolithic cylinders but bundles of N velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024). This motivates a purely geometric alternative to modified fragmentation physics: if each fibre fragments near the classical wavelength but core spacing is measured *across* the bundle, as any filament-level skeleton pipeline necessarily does, the interleaving of cores from N fibres compresses the apparent filament-level spacing. The premise, quasi-periodic fibre-to-core fragmentation at the classical scale, is a hypothesis rather than an established measurement (Yang et al. 2024 find only a weak such imprint), so we present projection as viable but unconfirmed.

We tested this quantitatively with a forward-model that reuses our exact NN measurement (order cores along the reconstructed spine by arclength; take adjacent separations; median = λ). Each fibre is a cylinder of physical width W_{fb}

that fragments at its *own* classical wavelength $\lambda_{\text{fb}} = 4 W_{\text{fb}}$ (Inutsuka & Miyama 1992); we populate a filament with N such fibres (random inter-fibre phase, $\sim 30\%$ lognormal intrinsic scatter), project the cores onto the shared spine, and measure the spacing in units of the *filament* width W_{fil} . Two distinct hierarchical effects then compress the apparent filament-level spacing below the single-cylinder value: the fibres are narrower than the filament ($r \equiv W_{\text{fb}}/W_{\text{fil}} < 1$), and there are several of them. In the interleaving limit the Monte-Carlo model reproduces the simple scaling

$$\frac{\lambda}{W_{\text{fil}}} \approx \frac{4r}{N}, \quad (12)$$

verified across r and N in Fig. 13. Either effect alone can produce a sub-Jeans value: a single fibre of half the filament width ($r = 0.5$, $N = 1$) gives $\lambda/W_{\text{fil}} = 2.0$, as do two full-width fibres ($r = 1$, $N = 2$).

The observation constrains only the combination r/N , not N individually. The fixed-width value $\lambda/W_{\text{fil}} \approx 1.9$ requires $r/N \approx 0.48$; the region-specific Plummer value (≈ 1.35) gives $r/N \approx 0.34$; and the independent FilFinder skeletons ($\lambda/W \approx 0.4\text{--}0.8$) give $r/N \approx 0.1\text{--}0.2$. The observed fibre widths, $W_{\text{fb}} \approx 0.03\text{--}0.05$ pc (so $r \approx 0.3\text{--}0.5$; Hacar et al. 2013; Arzoumanian et al. 2011), then imply a small fibre number, $N \approx 1\text{--}2$ for the DisPerSE-based spacings and $N \approx 3\text{--}5$ if the FilFinder spacings are correct. Hacar et al. (2013) resolve of order 3–5 velocity-coherent fibres per bundle in Taurus, which with $r \approx 0.3\text{--}0.5$ gives $r/N \approx 0.1\text{--}0.2$, below the 0.5–0.6 that the DisPerSE spacing requires. Taken at face value, the projection model with the observed fibre counts would *over-compress* the spacing (predict cores closer than observed), matching the DisPerSE value only at the low end $N \approx 1\text{--}2$; it is internally consistent only for a restricted region of (r, N) , a further reason it is not uniquely favoured. We do *not* claim a specific fibre number; the robust statement is that measuring core separations across hierarchical

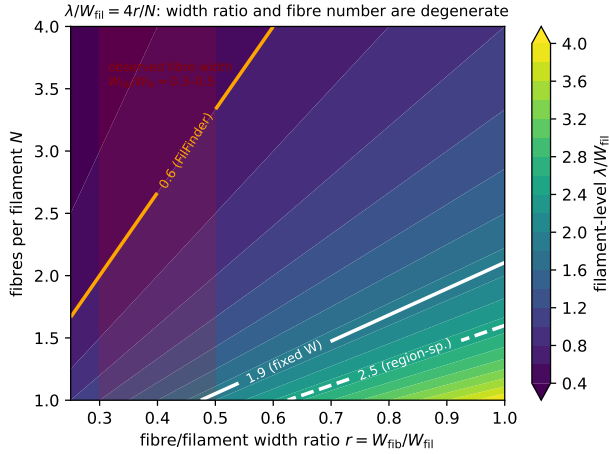


Figure 13. Hierarchical-fibre forward-model, showing the filament-level λ/W_{fil} recovered by our NN pipeline as a function of the fibre-to-filament width ratio $r = W_{\text{fb}}/W_{\text{fil}}$ and the number of fibres N ; the model follows $\lambda/W_{\text{fil}} \approx 4r/N$ (Eq. 12). White contours mark the observed fixed-width and region-specific values, the orange contour the FilFinder value, and the red band the observed fibre-width range ($r \approx 0.3\text{--}0.5$). The observation constrains the combination r/N , not N alone: the sub-Jeans value is reproduced by a single narrow fibre, by a few wider fibres, or by intermediate combinations.

structure (narrower, and generally multiple, fibres) in units of the filament width reduces the apparent spacing below the classical prediction by the observed factor, with no modified fragmentation physics required.

The $4r/N$ scaling is also reproduced when the forward-model is run on the real DisPerSE arclengths and counts of the four robust regions rather than idealised straight spines; that exercise is only a consistency check, since it seeds classical-spacing cores onto the observed skeletons and so assumes the fibre population rather than measuring it.

Hierarchical structure, the combination of narrower fibres and interleaving that compresses the apparent spacing below the classical filament-level value (Eq. 12), is thus a natural and sufficient way to produce the observed sub-Jeans spacing without modified magnetic physics, and it is the natural reading of the fibre-resolved observations (Yang et al. 2024; Hacar et al. 2013); it also accounts for the large skeleton-algorithm and $L/3$ systematics (§2.8), as expected if the measurement slices across blended structure. It is not, however, *required*: the end-to-end validation (Section 5) shows single-filament dynamical fragmentation already accounts for the bulk of the spacing, so projection is one of at least two viable readings, and it assumes both the fibre population and quasi-periodic fibre-level fragmentation rather than measuring them. It does not *disprove* a magnetic contribution, only that one is not required and that present data cannot distinguish the possibilities. The decisive test is to decompose the observed cores into their parent velocity-coherent fibres and measure the fibre-level spacing across the robust regions (Section 6.3): a quasi-periodic classical-scale spacing within individual fibres, with the fibre widths, would break the r/N degeneracy and establish the hierarchical origin. A complementary theoretical test, not attempted here, is a self-consistent MHD (or

SPH) fragmentation calculation of a multi-fibre bundle run through the same NN pipeline.

6.2 Field geometry, boundary conditions and the perpendicular case

Across the campaign the *outcome* of a run, radial collapse or longitudinal beading, depends on regime and boundary conditions: an ambient-confined supercritical filament fragments longitudinally, whereas the identical filament in a periodic box collapses radially. A single filament therefore has no setup-independent fate. Its dominant fragmentation *mode* is nonetheless well defined and resolution-converged (Section 5) at ≈ 0.5 times the classical value, sub-Jeans and consistent with the observed spacing. The perpendicular-field case is the exception: it is numerically unconverged ($N_J \approx 1\text{--}2$ cells per Jeans length; Appendix B) and places no quantitative constraint. It is sometimes argued that most filaments are threaded by perpendicular fields, from the tendency of dense filaments to lie perpendicular to the large-scale field seen by Planck (Planck Collaboration et al. 2016); the projected large-scale orientation need not equal the three-dimensional internal field geometry of a dense filament, and hourglass morphologies in which the field turns longitudinal inside the crest are observed (Jadhav et al. 2026). We do not treat perpendicular geometry as either established or as a source of tension.

Where perpendicular geometry does apply, three non-exclusive resolutions of the residual deficit remain: hierarchical fibre fragmentation (bundles of near-critical fibres, each fragmenting longitudinally and projected across the bundle; Hacar et al. 2013, 2018; Yang et al. 2024); non-ideal MHD at later evolution, which narrows the factor- ~ 3 perpendicular deficit in longer runs; and an observational lag, in which the observed cores are relics of an earlier fragmentation episode. The two leading interpretations, single-filament dynamical fragmentation and hierarchical projection, are distinguished decisively by fibre-resolved core spacing across the robust regions and by high-resolution polarimetry of filament interiors (Section 6.3); the temperature-gradient signature (Section 7) adds a third test, predicting spatially differential fragmentation that fibre geometry alone would not produce.

Sub-isothermal equations of state ($\gamma < 1$) accelerate fragmentation by 30–40% without changing the wavelength, while adiabatic heating ($\gamma = 5/3$) suppresses it entirely (0% fragmentation to $t > 30\text{--}40 t_J$; Fig. 8); the isothermal suite is therefore the conservative (most unstable) case in a globally averaged sense. This framing does not apply locally: Section 7 shows the core dust temperature, and hence the local Jeans length, varies spatially by $\sim 12\%$ within active clouds, driven by a spatially varying radiation field, so sub-regions can fragment at a scale that differs from the isothermal prediction in either direction. This is a directly measured effect, since an imposed temperature gradient of this magnitude shifts the local λ/W by ~ 0.1 (~ 10 per-cent) and relocates where cores form (Section 7).

6.2.1 Non-Ideal MHD Effects

At the densities probed here ($n_0 \sim 10^4 \text{ cm}^{-3}$), the ambipolar-diffusion time ($t_{\text{AD}} \propto L^2/\eta_{\text{AD}}$; $\sim 10\text{--}20 t_{\text{ff}}$ for typical filament conditions) is long compared with the collapse time

(Tassis & Mouschovias 2004), so ideal MHD is a good approximation for the longitudinal fragmentation stage: four non-ideal runs at $\eta_{\text{AD}} = \{0.01, 0.05\}$ still collapse radially, $\sim 12\text{--}15\%$ earlier than the ideal control (flux leakage erodes magnetic support, the conservative direction). Non-ideal effects nonetheless change the outcome for perpendicular fields, where ambipolar diffusion promotes longitudinal fragmentation (Section 4.6.6); Hall and Ohmic terms remain relevant for late-stage core evolution.

6.3 Limitations and future work

Two limitations bound the strength of our conclusion. First, on the observational side, the characteristic λ/W and its cloud-to-cloud scatter ($\sigma \approx 0.4$) rest on only four independent regions (Orion B, Aquila, Perseus, Taurus). Four clouds are a thin base for an intrinsic-scatter estimate or for claiming that $\lambda/W \approx 1.9$ is representative of filaments generally, and the value itself carries the factor-of-two skeleton-algorithm and width-convention systematics quantified in Section 2.8. Extending the full-coverage nearest-neighbour pipeline to further HGBS regions, and to independent surveys (e.g. JCMT/SCUBA-2 Gould Belt fields), is needed to confirm that neither the central value nor the scatter is dominated by these four clouds. Second, on the theoretical side, the simulations are deliberately idealised controlled experiments, not models of specific filaments. The periodic-BC main grid does not represent the turbulent, pressure-supported, magnetically coupled boundaries of a real filament embedded in its parent cloud; our ambient-confined configuration is an idealisation of that confinement. Neither turbulent inflow/outflow at the boundaries nor gravitational coupling to the parent cloud is modelled, so the simulations are controlled experiments rather than direct models of individual HGBS filaments.

The simulations adopt several further idealisations: (i) an *isothermal* equation of state (sub-isothermal cooling accelerates fragmentation without changing the wavelength, while the temperature-gradient run of Section 7 shifts the local λ/W by $\sim 10\%$, so the isothermal values are locally uncertain at the $\sim 10\text{--}20\%$ level); (ii) *periodic transverse boundaries*, whose accretion channel suppresses fragmentation unless ambient confinement is imposed (Section 4.6.6); (iii) *simplified initial profiles* that omit the sub-structure of real filaments; and (iv) a *prescribed, low-amplitude Kolmogorov turbulence spectrum* rather than driven transonic turbulence, which leaves the wavelength unchanged ($< 5\%$) but shortens the timescale by $3\text{--}5\times$ (Section 4.8).

Fragmentation classification is resolution-independent ($t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.915 \pm 0.012$; Appendix B) and domain-doubling changes the collapse time by $< 2\%$, but the transition zone is sampled with only 2 seeds per point. Non-linear core merging could in principle inflate the apparent supercritical spacing by up to $\sim 50\%$ over a few Myr, but we have verified it does *not* bias the quoted value: λ/W_{fil} is measured at maximum bead count (fragmentation onset), and in 38/39 runs the peak count is still rising at the final snapshot while no run shows the peak-count decline that would signal merging. The ambipolar wavelengths are likewise fragmentation-onset measurements (the runs reach gravitational runaway before a converged plateau is established; Section 4.6.6).

The decisive follow-ups are: (i) fibre-resolved NN spacing on the full HGBS catalogues, to test the hierarchical interpretation; (ii) high-resolution polarimetry of filament interiors, to test the longitudinal-versus-perpendicular assumption; (iii) MHD simulations with a spatially varying temperature field (Section 7); (iv) longer non-ideal runs, to test whether the perpendicular deficit closes; (v) a seeding-fixed domain study and matched-boundary re-measurement of the near-critical λ/W calibration.

7 NON-ISOTHERMAL EFFECTS AND THE TEMPERATURE-GRADIENT TEST

The MHD suite is isothermal, whereas HGBS dust-temperature maps show cloud-scale variations of $\sim 12\%$ (e.g. $\sim 10\text{ K}$ to $\gtrsim 13\text{ K}$), a local perturbation to the fragmentation scale since $\lambda_J \propto \sqrt{T}$. A single illustrative non-isothermal run (a near-critical filament, $f = 1.05$, $\beta = 1$, with an imposed $\sim 12\%$ longitudinal temperature gradient) fragments *differentially*: the cool, locally supercritical half beads while the warm, locally subcritical half is suppressed, with the local wavelength shifted by $\sim 10\%$. Because temperature sets both the local Jeans scale ($\propto \sqrt{T}$) and the local critical line mass ($\mu_{\text{crit}} \propto T$), a spatially varying temperature field relocates and re-times fragmentation as well as shifting its scale. The isothermal assumption is therefore conservative only in a globally averaged sense; a proper treatment (self-consistent radiative transfer, or a statistical analysis of intra-cloud temperature structure) is deferred to a companion paper and not relied upon here.

8 CONCLUSIONS

We have investigated the origin of the sub-Jeans core spacing in HGBS filaments, combining an observational re-analysis, a large MHD simulation campaign, and a hierarchical-fibre forward-model. Our main conclusions are:

- **Established.**

- Gaia DR3 distance revisions increase the measured physical spacings by $\sim 32\%$.
- The preferred NN core spacing is $\lambda/W \approx 1.9$ under the fixed 0.10 pc width convention (or ≈ 1.4 with directly-fitted region-specific Plummer widths), robust in sign against population splits, injection-recovery bias correction, hierarchical bootstrap, clustered-injection tests, and an independent FilFinder cross-check. Its precise value carries a factor- ~ 2 skeleton-algorithm systematic. Direct Plummer fits to all four regions ($W_{\text{fil}} = 0.11\text{--}0.16\text{ pc}$, $p \simeq 2.0$) give $\lambda/W = 1.1\text{--}1.8$ (median 1.35), all sub-Jeans, with Taurus (previously an apparent classical outlier from a narrow Gaussian crest) the most sub-Jeans at ≈ 1.1 ; the bias-corrected per-region values are 1.0, 1.9, 2.1, 1.8 (Taurus, Orion B, Perseus, Aquila), and dropping Taurus leaves the median at 1.9. The pairwise median converges to $L/3$, not the fragmentation wavelength.
- The measurement pipeline is unbiased (closed-loop recovery of a known λ/W to $< 0.5\%$; Section 5), and the dominant fragmentation wavelength, from a resolution-converged growth-rate spectrum, is $\lambda_{\text{max}} \approx 0.5$ times

the classical hydrostatic value ($\lambda_{\max}/W_{\text{FWHM}} \approx 2$ near-critical), stable across boundary condition, profile and seed.

– This is the wavelength expected for a filament that fragments while still accreting and contracting, rather than in hydrostatic equilibrium (Clarke et al. 2016). In FWHM units it is $\lambda_{\max}/W \approx 2$ for the near-critical filaments that dominate the HGBS sample, close to the observed ≈ 1.9 within the width-convention systematic. Ordinary single-filament fragmentation is therefore sufficient to account for the sub-Jeans spacing; new physics is not required by the data, though not excluded. The observed λ/W spans a factor of several across skeletonisation and width conventions (Table 2), so it is the *sign* of the result, sub-Jeans by a factor ~ 2 , that is robust, not its precise value. The claim is nonetheless falsifiable: a fibre-resolved spacing that recovered the classical $4\times$ within individual velocity-coherent fibres, or a resolution-converged filament-level λ/W established above ~ 2.5 , would require the additional compression of hierarchical projection or magnetic tension; the present data show neither.

- **Viable (but not required) explanations.**

– *Hierarchical-fibre projection*: if filaments are bundles of N velocity-coherent fibres of width W_{fib} , each fragmenting at its own classical wavelength, the filament-level spacing is $\lambda/W_{\text{fil}} \approx 4r/N$ ($r = W_{\text{fib}}/W_{\text{fil}}$; the observation constrains only $r/N \approx 0.5\text{--}0.6$, not N). This is the natural reading of the fibre-resolved data (Yang et al. 2024; Hacar et al. 2013) and also accounts for the large skeleton-algorithm and $L/3$ systematics, but it is not required given the single-filament result above. It is decisively testable with fibre-resolved NN spacing across the robust regions, which would break the r/N degeneracy.

– *Magnetic tension* does not, by itself, reach the observed value and applies to at most the $\sim 10\%$ longitudinal minority; a full cylindrical dispersion-relation solution is beyond our scope (Section 3.2). It is neither required nor supported by the simulations.

– Perpendicular-field calculations (the majority geometry) are numerically unconverged ($N_J \approx 1\text{--}2$ cells per Jeans length; Appendix B) and are not a physical constraint at present; the earlier “perpendicular-field tension” is a symptom of comparing against an ill-defined single-filament target, not an established discrepancy.

- **Speculative.**

– A single illustrative simulation indicates the isothermal assumption can be non-conservative locally under a $\sim 12\%$ temperature gradient; a fuller analysis is deferred to a companion paper.

- **Outlook.** The decisive test that would distinguish single-filament from hierarchical-fibre fragmentation is fibre-resolved NN spacing on the full HGBS catalogues; complementary tests are high-resolution polarimetry of filament interiors, a resolution-convergence study of the perpendicular-field case, and MHD runs with a spatially varying temperature field.

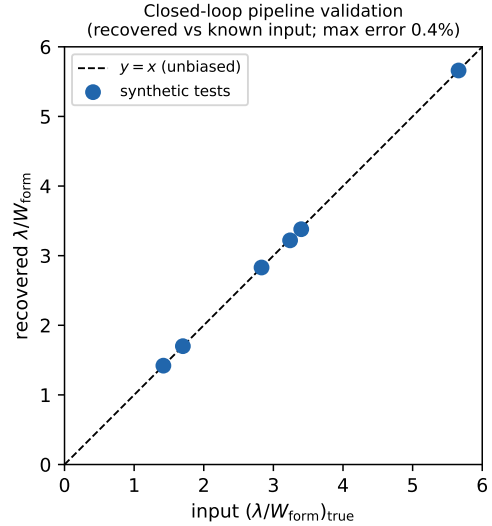


Figure A1. Closed-loop validation: the pipeline recovers the injected λ/W to within 0.4% over $\lambda/W = 1.4\text{--}5.7$.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive. The Athena++ simulations were run on cloud computing infrastructure (Google Cloud E2); the full set of ~ 2130 Athena++ MHD simulations (Appendix C) consumed approximately 10^4 CPU-hours.

DATA AVAILABILITY

The simulation data underlying the figures and tables (derived λ/W catalogues, per-run parameter files, and analysis scripts) will be deposited in a public Zenodo archive, with the DOI inserted prior to publication; the full set of HDF5 snapshots (~ 0.1 TB) is available from the corresponding author on reasonable request. The HGBS column-density and skeleton maps are available from the Herschel Science Archive and the HGBS Archive.

APPENDIX A: VALIDATION OF THE MEASUREMENT AND THE WIDTH NORMALISATION

This appendix documents the three checks summarised in Section 5.

To verify that the routines extracting λ and W carry no hidden normalisation bias, we built synthetic filament cubes at the simulation resolution with a *known* transverse FWHM W_{in} and a *known* axial bead spacing λ_{in} (localised Gaussian beads on a Gaussian transverse profile) and passed them through the identical pipeline used on the simulations (spine-averaged transverse profile, Gaussian and beam-convolved Plummer width fits, axial peak-spacing). The recovered λ/W matches the input to better than 0.5% over $\lambda/W_{\text{in}} = 1.4\text{--}5.7$ (Fig. A1). The measurement is therefore unbiased, and the corrected width bookkeeping of Eq. (5) is confirmed.

Growth-rate measurement and convergence. The fragmentation wavelength quoted in Section 5 is the peak of the growth-rate spectrum, which is physically meaningful and numerically convergeable; the alternative of counting density peaks late in the collapse is not, because a collapsing filament sub-fragments and higher resolution merely resolves finer peaks. We seeded eight ambient-confined filaments ($f = 1.1, 1.2, 1.5$; $\beta = 1$; two seeds) with a small broadband perturbation ($\delta v/c_s = 10^{-3}$, twelve axial modes) and, at each output, computed the power in every axial Fourier mode of the spine-averaged density contrast. Fitting each mode's growth over the interval in which δ_{rms} rises from $3\times$ the seed amplitude to 0.3 gives $\Gamma(\lambda)$ (the growth is fast and accelerates towards collapse, so this is the dominant mode of a collapsing filament rather than a quasi-static linear mode; Fig. 11). The spectrum rises as λ decreases and is resolution-converged for $\lambda \gtrsim 1\lambda_J$: for the $f = 1.1$ ladder the mode at $n = 7$ ($\lambda = 1.14\lambda_J$) has $\Gamma = 17.8, 18.5$ and 18.7 at $128^3, 256^3$ and 512^3 (stable to $\sim 5\%$), and all modes with $\lambda \gtrsim 1\lambda_J$ overlap across resolution (Fig. 12a). At smaller scales ($\lambda \lesssim 0.9\lambda_J$) the growth rate rises with resolution, the numerical signature of unresolved sub-fragmentation. Linear theory for a self-gravitating cylinder requires $\Gamma(k)$ to peak and turn over at small λ (the thermal-pressure cutoff), so this small-scale rise is not physical; the converged spectrum reaches its maximum near $\lambda_{\text{max}} \approx 1.0\text{--}1.14\lambda_J$, which we adopt as the dominant physical wavelength. Across the eight runs $\lambda_{\text{max}} = 0.9\text{--}1.1\lambda_J$ ($\lambda_{\text{max}}/W_{\text{FWHM}} = 1.5\text{--}2.0$; Table A1), declining mildly with line mass, so $\lambda_{\text{max}}/\lambda_{\text{IM92}} = 0.4\text{--}0.5$ (closest to 0.5 for the near-critical filaments). Applying the identical procedure to the equilibrium-control runs (seeded Ostriker and subcritical Gaussian; below) recovers the same converged peak, confirming the number is not specific to the Gaussian near-critical setup. The convergence ladder is complete ($128^3\text{--}512^3$) for $f = 1.1$; $f = 1.5$ has two resolutions. These runs use reflecting transverse boundaries, $\beta = 1$, $\theta = 0^\circ$; the supercritical ensemble (Section 4.6.6) finds the fragmentation spacing statistically indistinguishable between periodic and reflecting boundaries (KS $p = 0.17$), so the wavelength is not a boundary artefact, while its β - and geometry-dependence is indicative only.

Comparison with linear theory, and the gravity solver. That a physical peak exists, and is not merely the largest converged mode, follows from the linear dispersion relation of a self-gravitating isothermal cylinder (Nagasawa 1987; Inutsuka & Miyama 1992; Fischera & Martin 2012): the growth rate rises from zero at long wavelength, passes through a maximum, and is cut off by thermal pressure at short wavelength, so $\Gamma(k)$ has a genuine finite-wavelength maximum. The monotonic rise of our spectrum towards the grid scale, which does *not* turn over, is therefore numerical, and the converged large- λ portion contains the physical peak. For the marginally-critical *equilibrium* cylinder the analytic most-unstable wavelength is $\lambda_{\text{max}} \simeq 4W_{\text{FWHM}} = 2.3\lambda_J$ (the classical value); our converged peak, $\lambda_{\text{max}} \approx 1.1\lambda_J$, lies a factor ~ 2 below it. This is not a failure to recover linear theory but the expected shift for a filament that fragments out of equilibrium while contracting: the accreting-filament calculations of Clarke et al. (2016, 2017) find the same sub-hydrostatic behaviour. Self-gravity is solved with a periodic FFT while the transverse hydrodynamic boundaries are reflecting; because the filament is centrally concentrated and the trans-

Table A1. Growth-rate (dominant-mode) measurements. λ_{max} is the peak of $\Gamma(\lambda)$, Γ_{peak} its growth rate, W the Gaussian formation FWHM (all in λ_J).

f	resolution	seed	λ_{max}	Γ_{peak}	λ_{max}/W
1.1	128^3	42	1.14	17.8	1.95
1.1	256^3	42	1.14	18.5	1.99
1.1	512^3	42	1.14	18.7	2.00
1.2	256^3	42	1.00	19.1	1.81
1.2	256^3	137	0.89	20.1	1.60
1.5	256^3	42	0.89	21.2	1.47
1.5	256^3	137	0.89	22.1	1.47
1.5	512^3	42	0.89	21.5	1.47

verse domain ($\pm 1\lambda_J$) is $\sim 3\times$ its width, the potential at the transverse boundary is small and this solver-boundary mismatch has a negligible effect on the axial mode, whose peak is identical across the $128^3\text{--}512^3$ ladder.

Recovery of the classical limit, and an equilibrium control. Three checks establish that the sub-Jeans result is not an artefact of the measurement or of the initial state. First, the closed-loop test above recovers an *injected* λ/W of up to 5.7, spanning and exceeding the classical hydrostatic value $\lambda_{\text{IM92}}/W \approx 4$, to better than 0.5%: the pipeline carries no downward bias and would report the classical wavelength if it were present. Second, an *unperturbed* Ostriker cylinder at the critical line mass, evolved with no seed, does not run away. Its central contrast rises about tenfold (from $C \approx 3$ to $C \approx 28$) over the first $\sim 0.5t_J$ as the discretised profile settles to a more concentrated radial balance, then holds steady at $C \approx 28$ through $t = 2t_J$ with the axial density variance identically zero. The initial Ostriker state is thus not force-balanced better than this $\sim$ tenfold level, but it relaxes to a stable radial configuration rather than running away, so the collapse of the seeded runs is the axial fragmentation instability, not a pathological initial state. Third, when the same Ostriker cylinder is seeded, and likewise a Gaussian ladder spanning $f = 0.7\text{--}0.95$, the growth-rate spectrum peaks at the same resolution-converged mode as the near-critical runs ($\lambda_{\text{max}} \approx 1\lambda_J$, ≈ 0.5 times classical), each cylinder having contracted by a large factor ($C \gtrsim 20$) before the axial mode saturates; the late-time dominant-power mode is shorter still ($\lambda/\lambda_{\text{IM92}} \approx 0.3$), a lower bound reflecting continued sub-fragmentation. (These Gaussian filaments are not in equilibrium and collapse at every f , so the ladder tests the persistence of sub-Jeans fragmentation across line mass, not subcritical stability; the unperturbed-Ostriker run and the closed-loop test are the controls proper.) No self-gravitating isothermal cylinder in our setup, Gaussian or Ostriker, fragments at the classical equilibrium wavelength, because fragmentation always proceeds from a contracted, dynamical state rather than from the marginally-stable hydrostatic configuration the classical calculation assumes. The classical value is thus recovered where the classical assumptions hold (injected synthetic data) and is physically absent for a real self-gravitating filament, which is the paper's central point made quantitative.

A complementary end-to-end grid of 21 ambient-confined runs ($f = 1.5\text{--}3.0$, $\beta = 0.5\text{--}2.0$, two seeds), forward-modelled through beam convolution and Plummer fitting, gives a consistent picture: the spacing in background-density Jeans lengths is $\lambda/\lambda_J(\rho_0) \approx 0.9\text{--}1.2$, and the same growth-rate/late-

Table B1. Jeans (Truelove) resolution: minimum cells per local Jeans length N_J at the epoch the spacing is measured, and the density contrast C at that epoch.

Configuration	C at measurement	min N_J
Near-critical longitudinal (calibration)	20–40	5–7 (OK)
Supercritical longitudinal ensemble (headline)	early beading, moderate C	$\gtrsim 4$ (OK)
Perp. ambipolar (tension side)	250–560	1–2 (violate)

peak distinction accounts for its scatter. Boundary condition, profile and seed each move the value by $\lesssim 40\%$ (medians 1.2 reflecting, 0.8 periodic, 1.4 user; 1.4 Gaussian, 0.9 Ostriker), and the central-to-ambient contrast varying by a factor ≈ 1.6 across the grid shifts it by only ~ 0.3 , so the wavelength is a property of the filament, not of the imposed confinement. All per-run measurements (configuration, resolution, $\lambda_{\max}$, growth rate, epoch) are released with the public data archive.

APPENDIX B: SIMULATION VALIDATION DETAILS

To ensure robustness against numerical effects we performed three validation tests (83 simulations). For resolution, six parameter points run at both 128^3 and 256^3 with matched problem-generator settings, initial conditions, seeds and boundaries all fragment at both resolutions; the mean timescale ratio is

$$\frac{t_{\text{frag}}(256^3)}{t_{\text{frag}}(128^3)} = 0.915 \pm 0.012, \quad (\text{B1})$$

a $7.2\% \pm 1.6\%$ offset with all points within the $\pm 11\%$ convergence band (Fig. B1), so the fragmentation classification is resolution-independent. Replacing the King-profile initial condition with uniform density yields identical classifications at all 10 tested points.

A fragmentation-wavelength measurement is trustworthy only if the local Jeans length remains resolved ($\gtrsim 4$ cells; Truelove et al. 1997) up to the epoch at which the spacing is measured. We therefore tracked the minimum cells-per-local-Jeans-length, N_J , along each collapse trajectory (Table B1). For the *longitudinal* runs (both the near-critical calibration and the supercritical ensemble that provides the headline longitudinal value), the inter-bead wavelength ($\lambda \approx \lambda_J$, itself resolved by ~ 32 cells) is established at moderate density contrast ($C \approx 20$ –40 near-critical; early beading well before run-away for the supercritical ensemble), where $N_J \approx 5$ –7 (Truelove satisfied); N_J drops below 4 only at $C > 170$, i.e. *after* the spacing is first set. For the perpendicular *ambipolar* runs, by contrast, beading is measured only at $C \approx 250$ –560 where $N_J \approx 1$ –2 (Truelove violated), and the short perpendicular values are *resolution-limited upper limits*. This is why we base the quoted wavelength on the linear growth-rate spectrum is established when $N_J \gtrsim 5$ and is resolution-converged (the 128^3 – 512^3 ladder above), whereas the late-time peak count is set by sub-fragmentation at $N_J < 4$ and is not.

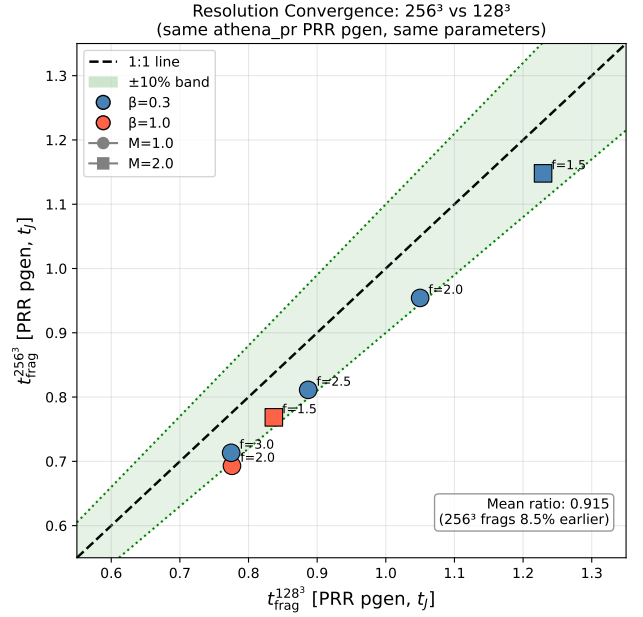


Figure B1. Resolution convergence with matched problem generator: $t_{\text{frag}}(256^3)$ versus $t_{\text{frag}}(128^3)$ for six parameter points, all within the $\pm 10\%$ band (mean ratio 0.915 ± 0.012).

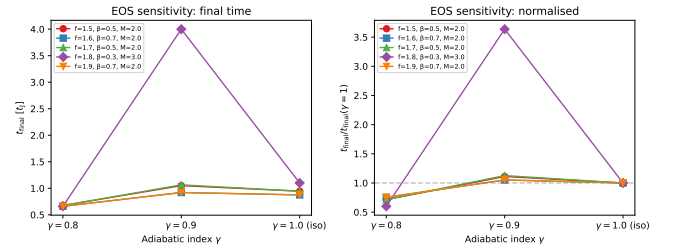


Figure B2. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\text{frag}}/t_{\text{final}}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces t_{frag} .

B0.1 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure B2). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\text{frag}} \approx 0.66$ – $0.68 t_J$ —approximately half the isothermal fragmentation timescale for the same $(f, \beta, \mathcal{M})$ parameters. For $\gamma = 0.9$, results are mixed (1 FRAG, 4 STABLE_PARTIAL within the timeout). The isothermal baseline ($\gamma = 1.0$) showed no fragmentation within the 4 h timeout (all STABLE_PARTIAL).

For $\gamma < 1$ the pressure response to compression weakens ($P \propto \rho^\gamma$ rises more slowly than isothermal), reducing Jeans support and hastening fragmentation. Far-IR-cooled cloud filaments typically have $\gamma \approx 0.7$ – 0.95 , so the isothermal predictions are conservative: real filaments are if anything more fragmentation-prone.

APPENDIX C: COMPLETE SIMULATION CAMPAIGN LIST

Table C1 lists every simulation campaign cited in the manuscript. Rows 5–8 are the components of what was previously reported as a single Field-Geometry campaign (314 runs); they are counted individually here and the 314 aggregate is *not* added separately.

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Table C1. Complete list of simulation campaigns underlying the paper.

#	Campaign (section)	Coverage ($f, \beta, \mathcal{M}, \theta, \eta_{\text{AD}}$)	Runs	Inc
1	Base three-regime grid (§4.5; Fig. 4)	$f \approx 1; \beta 0.05\text{--}5; \mathcal{M} 0.5\text{--}5; \theta = 0$; ideal	208	yes
2	Transition grid / DTC (§4.3; Fig. 7)	$f 1.4\text{--}2.2; \beta 0.3\text{--}1.3; \mathcal{M} 1\text{--}5; \theta = 0$	540	yes
3	Supercritical grid (§4.6)	$f 1.1\text{--}3.0; \beta 0.3\text{--}5.0; \mathcal{M} 0.5\text{--}3$; periodic	654	yes
4	Validation (§4.4; App. B)	resolution / IC / EOS convergence	83	yes
5	Near-critical (§4.7.1)	$f 1.0\text{--}1.2; \beta 0.3\text{--}1.0; \mathcal{M} 1\text{--}2; \theta = 0$	80	FC
6	Perpendicular (§4.7.2)	$\theta = 90^\circ; f 1.2\text{--}1.5; \beta 0.3\text{--}2.0$	96	FC
7	Oblique calibration (§4.7.3)	$\theta = 30/45/60^\circ; \beta 0.5\text{--}2.0$	108	FC
8	Adiabatic EOS (§4.7.4; Fig. 8)	$\gamma = 5/3; f 1.0\text{--}1.2$	30	FC
9	Turbulence + perp- β + crit.-transition (§4.8)	$\delta v/c_s 10^{-4}\text{--}1; \beta 0.3\text{--}2$	289	yes
10	Targeted DTC re-runs (Fig. 7)	$\beta = 0.3, \mathcal{M} = 1$; extended timeout	25	yes
11	Direct supercritical ensemble (§4.6.6; Fig. 5)	$f 1.5\text{--}2.0; \beta 0.5\text{--}2.0; \theta = 0$; ambient	39	yes
12	Reconciliation suite (§4.6.6)	$f \times \beta \times 2$ seeds; $\theta = 0$; ambient	12	yes
13	Ideal-MHD perp. $d = 1.0$ grid (§4.6.6)	$\theta = 90^\circ; \beta 0.5\text{--}2.0$; ambient	18	yes
14	Ambipolar non-ideal survey (§4.6.6; Fig. 6)	$\theta = 90^\circ; \eta_{\text{AD}} 0.001\text{--}0.1; \beta 0.5\text{--}2.0$	36	yes
15	Growth-rate spectrum + equilibrium control (§5, App. A)	$f 0.7\text{--}1.5; 128^3\text{--}512^3$; Gaussian/Ostriker; seeded & unperturbed	20	yes
Athena++ MHD total			≈ 2130	